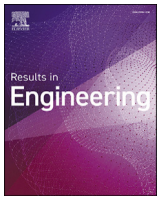
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Research paper



Development and numerical validations of a two-dimensional cumulant-driven thermal lattice Boltzmann method

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ABSTRACT

A novel cumulant-based lattice Boltzmann method (CuLBM) has been developed for the two-dimensional (2D) D2Q9 lattice model to analyze velocity and the D2Q5 lattice model for temperature fields. The computer code has been bench-marked for natural convection in fluid flow and heat transfer within a side-heated square cavity for the Newtonian and non-Newtonian power-law and Bingham fluids, utilizing compute unified device architecture (CUDA) C/C++ through graphics processing unit (GPU) computing. This study details the mathematical formulation of the cumulant-based 2D lattice Boltzmann method for the velocity and passive scalar fields. A wide range of flow Rayleigh numbers, from $Ra = 10^3$ to $Ra = 10^9$, has been employed for the Newtonian fluids, for the non-Newtonian fluids the maximum $Ra = 10^6$ for the various power-law index $n = 0.6$ to 1.4 and the Bingham numbers $Bn = 0$ to 15, and results are presented in terms of streamlines, isotherms, velocity, temperature, local and average rate of heat transfer, and flow entropy generation. These validations demonstrate that the current two-dimensional CuLBM will be a superior option for solving convective heat transfer problems for the Newtonian and non-Newtonian complex fluids.

1. Introduction

The Lattice Boltzmann method (LBM) is widely recognized as an effective alternative for solving equations like the Navier-Stokes or energy equations, based on the lattice Boltzmann equation. The most commonly used variant of LBM is the single-relaxation-time (SRT) or Bhatnagar-Gross-Krook (BGK) LBM. However, a significant limitation of the BGK method is its reduced numerical stability at higher Reynolds numbers (Re) or Rayleigh numbers (Ra), where dynamic viscosity or thermal diffusivity is notably low. This issue was addressed with the introduction of the raw-moments-based multiple-relaxation-time (MRT) LBM, which offers improved stability compared to the BGK-LBM. Nevertheless, challenges remain for extremely low viscosity or thermal diffusivity due to insufficient Galilean invariance.

To tackle these issues, Geier et al. [1] identified weaknesses in the raw-moments-based MRT approach and proposed a new three-dimensional (3D) D3Q27 lattice model that employs a central moments-based cascaded lattice Boltzmann method (CLBM). This innovative technique significantly enhances numerical stability for flows characterized by low viscosity or thermal diffusivity. Since then, numerous studies have explored 2D and 3D fluid flow problems using the cascaded lattice Boltzmann method [2–5].

Geier et al. [6] also introduced a new 3D D3Q27 lattice Boltzmann method (LBM) that utilizes a statistical cumulant collision operator instead of the conventional moments operator. They argued that cumulants offer improved Galilean invariance and a more effective decoupling of independent degrees of freedom compared to the moments-based approach. The following paragraphs will provide a brief overview of the literature related to cumulant-based LBM (CuLBM).

In their work, Geier et al. [7,8] presented a comprehensive derivation of the D3Q27 cumulant LBM and explored its applications, showing that CuLBM achieves superior convergence with fourth-order accuracy compared to other lattice Boltzmann techniques. Far et al. [9] utilized the D3Q27 CuLBM to model the dispersion of ceramic agglomerates, while Kutscher et al. [10] applied the large-eddy simulation (LES) cumulant LBM for multiscale simulations of turbulent flow interacting with porous media, achieving excellent correlation with both experimental and direct numerical simulation (DNS) data. Eichler et al. [11] investigated turbulent flows over rough surfaces using CuLBM as a direct numerical solver (DNS), and their findings were consistent with

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results from finite difference methods and wind tunnel experiments. Gehrke and Rung [12] simulated turbulent channel flow using LES-based CuLBM at a maximum Reynolds number of $Re = 50,000$, finding strong agreement with DNS data. Furthermore, Sato et al. [13] employed LES-based CuLBM to study single-phase free surface models in coastal and ocean engineering flows.

Sitompul et al. [14] explored violent two-phase flows using CuLBM, successfully simulating phenomena such as Rayleigh-Taylor instability, rising bubbles, and droplet dynamics. Maeyama et al. [15] applied CuLBM to investigate peristaltic laminar blood flow in a stenotic artery, utilizing a non-Newtonian power-law model for low Reynolds numbers ($Re \leq 10$) with a power-law index of ($n = 0.21$). Adekanye et al. [16] studied passive scalar dilute gravity currents with GPU-accelerated CuLBM, comparing their D3Q27 CuLBM results to those obtained from the D3Q19 BGK-LBM. They concluded that CuLBM serves as a more effective numerical tool for accurately capturing gravity current flows.

The three-dimensional (3D) D3Q27 CuLBM has already shown to be more robust compared to methods like the Bhatnagar-Gross-Krook (BGK), single relaxation-time (SRT), two-relaxation-time (TRT), raw-moment-based multiple-relaxation-time (MRT), and central moments-based cascaded lattice Boltzmann method (CLBM). Previous studies have primarily concentrated on the D3Q27 lattice model in the CuLBM context. To our knowledge, only one study by Venturi et al. [17] applied a two-dimensional (2D) cumulant LBM to solve the shallow water equations. However, their work could have provided a comprehensive, step-by-step formulation of the 2D CuLBM, leaving room for a more detailed explanation.

Natural convection in a square cavity has long been a classic benchmark problem for testing numerical methods in fluid dynamics. This is largely due to its clear-cut geometry and its sensitivity to various thermal and flow parameters, particularly the Rayleigh number. One of the groundbreaking studies in this area was conducted by de Vahl Davis [18] in 1983, who provided a benchmark numerical solution for laminar natural convection of air in a square cavity that was heated from the side. Using the finite difference method along with the streamfunction-vorticity formulation, this research yielded highly accurate results for Rayleigh numbers ranging from 10^3 to 10^6. These findings quickly became the go-to reference solutions for validating the accuracy of new numerical schemes. Building on this foundational work, Le Quere [19] in 1991 pushed the boundaries further by enhancing the accuracy of numerical solutions at even higher Rayleigh numbers, reaching up to 10^8. He achieved this by employing advanced numerical discretization and grid resolution techniques. His research underscored the importance of meticulous mesh refinement and numerical stability when simulating flows at high Rayleigh numbers, where boundary layer effects start to play a significant role.

As mesoscopic approaches evolved, Dixit and Babu [20] utilized the Lattice Boltzmann Method (LBM) to simulate natural convection in a square cavity at high Rayleigh numbers, going up to 10^9. Their study showcased LBM's ability to effectively capture essential flow features and heat transfer characteristics while maintaining reasonable computational efficiency and accuracy. They also stressed the significance of proper boundary conditions and ensuring grid independence in LBM simulations for thermal flows. Liu et al. [21] introduced a non-orthogonal multiple-relaxation-time (MRT) lattice Boltzmann method (LBM) designed for incompressible thermal flows. This innovative framework enhances numerical stability and accuracy compared to traditional single-relaxation-time and orthogonal MRT methods [21]. Their approach shines when tackling complex geometries and high Rayleigh number scenarios, making it a valuable asset for real-world engineering tasks that involve natural convection. Sharma et al. [22] contributed significantly to this field by introducing the cascaded thermal Lattice Boltzmann Method (CTLBM) for simulating natural convection heat transfer. The CTLBM leverages a central-moment-based collision model, which enhances numerical stability and accuracy over standard BGK and MRT models. The authors demonstrated that the cascaded formulation reduces numerical diffusion and better captures the temperature and velocity fields, especially in thermally driven flows within square cavities. Their results aligned well with benchmark solutions, highlighting the method's potential for complex thermal flow modeling. Expanding on the LBM framework, Nee evaluated hybrid Lattice Boltzmann models for both laminar and turbulent natural convection regimes [23]. The study involved coupling LBM with traditional turbulence modeling techniques to handle a broader range of Rayleigh numbers, up to the turbulent regime. Nee's results showcased the strengths and limitations of different hybrid strategies, such as LES-LBM and hybrid MRT-LES models, in capturing flow structures and heat transfer characteristics across varying regimes. This work is particularly relevant for extending LBM's applicability to high-Rayleigh-number flows, which are often characterized by complex instabilities and turbulence.

On another note, Trias et al. [24] took a Direct Numerical Simulation (DNS) approach to explore natural convection in a differentially heated cavity with an aspect ratio of 4 and Rayleigh numbers soaring as high as 10^{11}. This research stands out for its high-resolution analysis of turbulent convection, where they managed to resolve all relevant spatial and temporal scales without relying on modeling approximations. In Part I of their study, they concentrated on the numerical formulation and provided detailed time-averaged flow structures, which serve as valuable benchmark data for validating turbulence models and numerical methods. Their DNS findings highlighted crucial aspects of high-Rayleigh-number convection, such as boundary layer dynamics and plume formation, both of which are essential for making accurate thermal predictions in engineering applications.

Collectively, these studies illustrate a clear progression in computational methods for modeling natural convection, evolving from improved LBM formulations with central moments to hybrid turbulence models and comprehensive DNS. Each approach offers distinct advantages based on the target Rayleigh number, available computational resources, and the level of accuracy required. The recent trend suggests a shift towards utilizing mesoscopic LBM frameworks for engineering efficiency, alongside DNS studies that provide deeper insights into fundamental flow physics.

A review of the literature reveals that there has been no research conducted using thermal CuLBM for the passive scalar fields with Newtonian and non-Newtonian fluids. In this study, we present a D2Q9 lattice model for the velocity field and a D2Q5 lattice model for the temperature field, providing a step-by-step validation of the GPU-accelerated CUDA C++ code through simulations of natural convection in fluid flow, heat transfer, and entropy production within a side-heated cavity flow. This comprehensive formulation aims to support the LBM community in easily developing their own in-house codes for various applications. Natural convection in a side-heated cavity is an essential heat transfer process where fluid movement is driven by buoyancy forces resulting from temperature differences along vertical walls. In this setup, one vertical wall is heated (at a high temperature), while the opposite wall is cooled (at a low temperature), creating a temperature gradient that leads to fluid circulation. This phenomenon is important in various engineering and environmental contexts. In practical applications, side-heated cavities are often seen in building insulation, cooling of electronic devices, solar collectors, and industrial heat exchangers. Gaining a deeper understanding of the heat transfer and fluid flow characteristics in these configurations is key to optimizing energy efficiency and enhancing system performance in thermal management applications. Computational simulations research on natural convection for the Newtonian and non-Newtonian fluids offer valuable insights for improving these systems' efficiency and sustainability.

2. Benchmark problem statement

The benchmark problem for validating the current Cumulant Lattice Boltzmann Method (CuLBM) code focuses on natural convection within a two-dimensional (2D) square cavity with Newtonian and non-

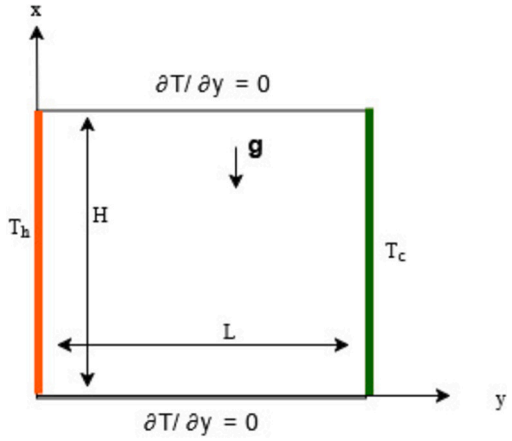


Fig. 1. Computational domain and coordinate systems.

Newtonian fluids. This scenario is a common test case in computational fluid dynamics (CFD) used to assess the accuracy and stability of numerical methods for simulating convective heat transfer in enclosed spaces. The computational domain, depicted in Fig. 1, features a square cavity with a height H and a length L. The cavity is heated differentially, meaning that temperature differences along the walls drive the convective flow within the enclosure. The specific boundary conditions for this problem are outlined as follows:

- Left Wall (Hot Wall): The left vertical wall is kept at a constant high temperature T_h, providing a steady heat source along this boundary.
- Right Wall (Cold Wall): The right vertical wall is maintained at a lower temperature T_c, establishing a temperature gradient across the cavity.
- Top and Bottom Walls (Adiabatic Walls): The horizontal walls are thermally insulated, indicating that no heat transfer occurs through these boundaries ($\frac{\partial T}{\partial y} = 0$).
- Gravity Effect: The influence of gravity g is taken into account in the downward direction, affecting the buoyancy-driven flow within the cavity.

To validate the accuracy and robustness of the developed CuLBM code, simulations are conducted across a broad range of Rayleigh numbers ($Ra = 10^3$ to 10^9)for the Newtonian fluids, for the non-Newtonian fluids the maximum $Ra = 10^6$ for the various power-law index $n = 0.6$ to 1.4 and the Bingham numbers $Bn = 0$ to 15. The results are compared with established numerical data by examining:

- Streamlines and Velocity Field: To observe the development of convective flow patterns.
- Isotherms: To analyze thermal stratification and heat transfer.
- Flow and Thermal Characteristics: Including velocity, temperature, and temperature gradients.
- Entropy Generation: To assess the efficiency of heat transfer and the irreversibilities within the system.

The square cavity problem serves as a standard test case in CFD due to its:

- Simple yet rich physical setup for validating numerical methods.
- Direct comparison with available numerical and experimental results.
- Controlled environment for evaluating numerical stability and convergence.

3. Governing equation

In the context of two-dimensional (2D) laminar and incompressible fluid flow and heat transfer the governing equations are the Navier-Stokes and energy equations under the Boussinesq approximation. These equations can be expressed in the following dimensionless form as like [25–27]:

Continuity equation

$$\frac{\partial \overline{u}}{\partial \overline{x}} + \frac{\partial \overline{v}}{\partial \overline{y}} = 0 \tag{1}$$

u-momentum equation

$$\rho\left(\frac{\partial \overline{u}}{\partial \overline{t}} + \overline{u}\frac{\partial \overline{u}}{\partial \overline{x}} + \overline{v}\frac{\partial \overline{u}}{\partial \overline{y}}\right) = -\frac{\partial \overline{p}}{\partial \overline{x}} + \frac{\partial}{\partial \overline{x}}\left(2\mu\frac{\partial \overline{u}}{\partial \overline{x}}\right) + \frac{\partial}{\partial \overline{y}}\left(\mu\frac{\partial \overline{u}}{\partial \overline{y}}\right) + \frac{\partial}{\partial \overline{y}}\left(\mu\frac{\partial \overline{v}}{\partial \overline{x}}\right) \tag{2}$$

v-momentum equation

$$\rho\left(\frac{\partial \overline{v}}{\partial \overline{t}} + \overline{u}\frac{\partial \overline{v}}{\partial \overline{x}} + \overline{v}\frac{\partial \overline{v}}{\partial \overline{y}}\right) = -\frac{\partial \overline{p}}{\partial \overline{y}} + \frac{\partial}{\partial \overline{x}}\left(\mu\frac{\partial \overline{v}}{\partial \overline{x}}\right) + \frac{\partial}{\partial \overline{y}}\left(2\mu\frac{\partial \overline{v}}{\partial \overline{y}}\right) + \frac{\partial}{\partial \overline{x}}\left(\mu\frac{\partial \overline{u}}{\partial \overline{y}}\right) + g(\rho\beta)\left(T - T_c\right) \tag{3}$$

Energy equation

$$\frac{\partial T}{\partial \overline{t}} + \overline{u}\frac{\partial T}{\partial \overline{x}} + \overline{v}\frac{\partial T}{\partial \overline{y}} = \alpha\left(\frac{\partial^2 T}{\partial \overline{x}^2} + \frac{\partial^2 T}{\partial \overline{y}^2}\right) \tag{4}$$

where $\overline{u}$ and $\overline{v}$ are the fluid velocity components, T is the dimensional fluid temperature, μ denotes the Newtonian molecular viscosity or non-Newtonian viscosity and ρ is the density of the fluid, g is the acceleration due to the gravity, β is the thermal expansion coefficient, and α is the thermal diffusivity of fluid.

There are few important non-dimensional numbers which are related to the natural convection problem as: $Ra \overset{\text{def}}{=} \frac{g\beta\Delta T H^3}{\alpha\nu}$ is the Rayleigh number and $Pr \overset{\text{def}}{=} \frac{\nu}{\alpha}$ is the Prandtl number, $U = \frac{\alpha}{H}$ is the characteristic velocity. Here H is the characteristic length, ν is the kinematic viscosity,

4. Formulation of the D2Q9 cumulant lattice Boltzmann method (CuLBM)

When developing a cumulant-based Lattice Boltzmann Method (LBM) to simulate incompressible thermal flows, choosing the right lattice models for both the velocity and temperature fields is essential for achieving accuracy, stability, and computational efficiency. In this study, we use the D2Q9 lattice for the velocity field and the D2Q5 lattice for the temperature field, guided by several important considerations:

The D2Q9 lattice model is the go-to two-dimensional setup in Lattice Boltzmann Method (LBM) for simulating isothermal and weakly compressible flows. Its application within a cumulant-based framework is particularly well-founded for several reasons:
(i) The D2Q9 model showcases fourth-order isotropy, which is crucial for accurately recovering the Navier–Stokes equations in the continuum limit. This characteristic ensures that the viscous stress tensor is represented correctly in every direction.
(ii) The cumulant-based LBM functions in moment space, converting distribution functions into cumulants that represent central moments of increasing order. The D2Q9 model offers enough degrees of freedom to articulate these cumulants (like zeroth-order mass, first-order momentum, and second-order stresses), allowing for precise control over the relaxation of each cumulant mode, which in turn enhances numerical stability and minimizes non-physical artifacts.
(iii) By applying distinct relaxation rates to various cumulants, the

cumulant-based approach using D2Q9 significantly boosts stability at high Reynolds numbers or Rayleigh numbers and improves Galilean invariance, particularly in low-viscosity scenarios.
(iv) The D2Q9 model is compatible with a broad range of well-established boundary schemes (such as bounce-back and Zou–He), which can be easily extended or adapted within the cumulant framework.

The D2Q5 lattice is a handy tool for tackling the advection-diffusion equation when looking at temperature fields, especially under the passive scalar or Boussinesq approximation. Here's why it's a solid choice within the cumulant-based LBM framework:
(i) The D2Q5 model does a great job of capturing the thermal advection-diffusion dynamics with fewer degrees of freedom, which means it doesn't have to deal with the complexities of momentum transport. You can represent the scalar temperature field using low-order cumulants (like zeroth and second order), which are just right for this task.
(ii) Just like D2Q9, you can apply cumulant relaxation to D2Q5, which enhances stability and allows for adjustable thermal diffusivity. This is particularly useful when simulating flows with high Prandtl numbers or sharp thermal gradients.
(iii) D2Q5 has fewer discrete velocities compared to D2Q9, which results in a smaller memory footprint and quicker computations for the temperature solver, all while keeping the accuracy of scalar transport intact.
(iv) Since temperature influences the flow field solely through a buoyancy force term (as per the Boussinesq approximation), a simpler lattice like D2Q5 is perfectly capable of representing temperature changes while still accurately coupling with the D2Q9 velocity field.

The combination of D2Q9 for velocity and D2Q5 for temperature within the cumulant-based LBM framework strikes a great balance between physical accuracy and computational efficiency. The higher-order isotropy and expressive nature of D2Q9 make it perfect for modeling fluid momentum dynamics accurately. Meanwhile, the lightweight yet effective structure of D2Q5 allows for efficient and stable simulations of the thermal field as a passive scalar. This duo is especially well-suited for tackling incompressible thermal convection problems that follow the Boussinesq approximation.

Geier et al. [6] were the first to introduce the cumulant-based lattice Boltzmann method for three-dimensional (3D) flows utilizing the D3Q27 lattice model, demonstrating that it offers superior numerical stability and degrees of freedom compared to other existing lattice Boltzmann methods. In this work, we present a step-by-step description of the 2D thermal cumulant lattice Boltzmann method (CuLBM) and validate it for the natural convection in a side heated square cavity.

In probability theory and statistics, cumulants are a set of quantities that provide an alternative means of characterizing a probability distribution, distinct from its moments. It is well established that the Navier-Stokes equations (1)-(4) can be solved using various lattice Boltzmann methods. However, the numerical stability and robustness of these methods can vary significantly. The discretized time-dependent lattice Boltzmann equation can be expressed as follows [27]:

$$f_i(\mathbf{r}+\mathbf{q_i}\Delta t, t+\Delta t) - f_i(\mathbf{r},t) = \Omega_i \tag{5}$$

Here, f_i and Ω_i represent the discrete momentum distribution functions and the collision operator, respectively. The vector $\mathbf{r}$ is defined as (x, y). The standard parameters are set with lattice spacing $\Delta x = \Delta y$, a time step Δt, and a lattice speed given by $c = \frac{\Delta x}{\Delta t} = 1$. The discrete velocities $\mathbf{q_i} = \left[|\xi_i\rangle, |\eta_j\rangle\right]$ are defined as follows [27]:

$$|\xi_i\rangle = \left[\,0, \quad 1, \quad 0, \quad -1, \quad 0, \quad 1, \quad -1, \quad -1, \quad 1\,\right]^T, \tag{6}$$

$$|\eta_j\rangle = \left[\,0, \quad 0, \quad 1, \quad 0, \quad -1, \quad 1, \quad 1, \quad -1, \quad -1\,\right]^T, \tag{7}$$

Here, the column vector is indexed by $i, j = 0, \ldots, 8$, while the superscript T indicates transposition. The collision step is performed within the cumulant space of the collision operator.

The lattice Boltzmann method primarily consists of two steps: streaming and collision. The momentum distribution functions are transferred to neighboring nodes during the streaming step. In contrast, in the collision step, the cumulants of the local distribution functions are computed and relaxed toward their equilibrium states.

In the equilibrium state, all degrees of freedom must be statistically independent. The joint probability of these degrees of freedom can be viewed as the product of the probabilities for each independent degree. The Maxwell distribution function exemplifies this property. The Maxwell distribution represents the equilibrium distribution, which is both continuous and differentiable [6,28].

A detailed description of the cumulant formation for the D3Q27 CuLBM for the velocity field was given in two PhD theses [29,30]. Based on these theses, the cumulants for the D2Q9 and D2Q5 models will be derived and presented as follows:

The moment generating function of the Maxwell distribution in two-dimensions (2D) is:

$$M^{eq}(\mathcal{X},\mathcal{Y}) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f^{eq}(\xi,\eta)e^{\mathcal{X}\xi}e^{\mathcal{Y}\eta}d\xi d\eta \tag{8}$$

where the Maxwell equilibrium distribution function is:

$$f^{eq}(\xi,\eta) = \rho\left(\frac{1}{2\pi C_s^2}\right)^{3/2} e^{-\frac{(c\xi-u)^2+(c\eta-v)^2}{2C_s^2}} \tag{9}$$

The equilibrium function (9) is substituted into the moment generating function (8), yields:

$$M^{eq}(\mathcal{X},\mathcal{Y}) = \rho\left(\frac{1}{2\pi C_s^2}\right)^{3/2} \int_{-\infty}^{\infty} e^{-\frac{(c\xi-u)^2}{2C_s^2}+\mathcal{X}\xi} d\xi \int_{-\infty}^{\infty} e^{-\frac{(c\eta-v)^2}{2C_s^2}+\mathcal{Y}\eta} d\eta \tag{10}$$

After integration and simplification, a normalized Maxwell distribution function is:

$$M^{eq}(\mathcal{X},\mathcal{Y}) = e^{u\mathcal{X}/c+v\mathcal{Y}/c+\frac{C_s^2(\mathcal{X}^2+\mathcal{Y})^2}{2c^2}} \tag{11}$$

where $\mathcal{X}$ and $\mathcal{Y}$ are the normalized wave numbers.

For the 2D case a probability density function (PDF) for the raw and central moments can be defined as, respectively:

$$f_r(\xi,\eta) = \sum_{ij} f(\xi_i,\eta_j)\delta(\xi)\delta(\eta) \tag{12}$$

$$f(\xi,\eta) = \sum_{ij} f(\xi_i,\eta_j)\delta(\xi-(\xi_i-u))\delta(\eta-(\eta_j-v)) \tag{13}$$

The moments for this PDF can be calculated as:

$$\mu_{\xi^m\eta^n} = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f_r(\xi,\eta)\xi^m\eta^n d\xi d\eta \tag{14}$$

The central moments for this PDF can be calculated as:

$$K_{\xi^m\eta^n} = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(\xi,\eta)\xi^m\eta^n d\xi d\eta \tag{15}$$

Using the equation (8) into (9), yields

$$K_{\xi^m\eta^n} = \sum_{ij} f(\xi_i,\eta_j)(\xi_i-u)^m(\eta_j-v)^n = \sum_{ij} f(\xi_i,\eta_j)(q_{i\xi}-u)^m(q_{i\eta}-v)^n \tag{16}$$

The central moment generating function is acquired the same way as the raw moment generating function, as:

$$\hat{M}(\mathcal{X},\mathcal{Y}) = e^{-u\mathcal{X}/c-v\mathcal{Y}/c}\sum_{ij} f(\xi_i,\eta_j)e^{\mathcal{X}\xi}e^{\mathcal{Y}\eta} = e^{-u\mathcal{X}/c-v\mathcal{Y}/c}M(\mathcal{X},\mathcal{Y}) \tag{17}$$

Therefore, central moments can be calculated directly as:

$$K_{\xi^m\eta^n} = \frac{\partial^m}{\partial\mathcal{X}^m}\frac{\partial^n}{\partial\mathcal{Y}^n}\hat{M}(\mathcal{X},\mathcal{Y})|_{\mathcal{X}=\mathcal{Y}=0} \tag{18}$$

The cumulant generating function is

$$C_{\xi^m\eta^n} = \frac{\partial^m}{\partial\mathcal{X}^m}\frac{\partial^n}{\partial\mathcal{Y}^n}\ln\left(M(\mathcal{X},\mathcal{Y})\right)|_{\mathcal{X}=\mathcal{Y}=0} \tag{19}$$

Each cumulant have its own relaxation rate $\omega_{\xi^m\eta^n}$ as used the post-collision state,

$$C^*_{\xi^m\eta^n} = C_{\xi^m\eta^n} + \omega_{\xi^m\eta^n}\left(C^{eq}_{\xi^m\eta^n} - C_{\xi^m\eta^n}\right) \tag{20}$$

where $C^{eq}_{\xi^m\eta^n}$ is the equilibrium cumulant.

4.1. D2Q9 momentum cumulant lattices Boltzmann method

For the D2Q9 lattice model, the central moments can be calculated as follows:

$$\begin{aligned}
&K_{00} = \rho = \sum_i f_i, \quad K_{10} = \sum_i(\xi_i-u)f_i = 0,\\
&K_{01} = \sum_i(\eta_i-v)f_i = 0, K_{20} = \sum_i(\xi_i-u)^2 f_i,\\
&K_{02} = \sum_i(\eta_i-v)^2 f_i,\\
&K_{11} = \sum_i(\xi_i-u)(\eta_i-v)f_i
\end{aligned} \tag{21}$$

Now normalize the central moments by the ρ, equation (21) becomes as:

$$\begin{aligned}
&\hat{K}_{00} = 1, \quad \hat{K}_{10} = K_{10}/\rho, \quad \hat{K}_{01} = K_{01}/\rho\\
&\hat{K}_{20} = K_{20}/\rho, \quad \hat{K}_{02} = K_{02}/\rho, \quad \hat{K}_{11} = K_{11}/\rho\\
&\hat{K}_{21} = K_{21}/\rho, \quad \hat{K}_{12} = K_{12}/\rho, \quad \hat{K}_{22} = K_2/\rho
\end{aligned} \tag{22}$$

The details of the cumulant calculations for the D3Q27 CuLBM were shown step by step in the PhD thesis of Fard [29]. Following this thesis, the present cumulants are calculated for the D2Q9 CuLBM as:

$$\begin{aligned}
&C_{00} = \ln(\hat{K}_{00}) = 0, \quad C_{10} = \hat{K}_{10}, \quad C_{01} = \hat{K}_{01},\\
&C_{20} = \hat{K}_{20}, C_{02} = \hat{K}_{02}, \quad C_{11} = \hat{K}_{11},\\
&C_{21} = \hat{K}_{21}, \quad C_{12} = \hat{K}_{12}\\
&C_{22} = \hat{K}_{22} - \hat{K}_{20}\hat{K}_{02} - 2\hat{K}_{11}\hat{K}_{11}
\end{aligned} \tag{23}$$

With the help of equations (11) and (19), the cumulant equilibrium distribution functions are:

$$C^{eq}_{00} = \rho, C^{eq}_{10} = C^{eq}_{01} = 0, C^{eq}_{20} = C^{eq}_{02} = \rho C_s^2, \quad C^{eq}_{11} = C^{eq}_{21} = C^{eq}_{12} = C^{eq}_{22} = 0 \tag{24}$$

The normalized cumulant equilibrium distribution functions are:

$$C^{n\,eq}_{00} = 1, C^{n\,eq}_{10} = C^{n\,eq}_{01} = 0, C^{n\,eq}_{20} = C^{n\,eq}_{02} = C_s^2, \quad C^{n\,eq}_{11} = C^{n\,eq}_{21} = C^{n\,eq}_{12} = C^{n\,eq}_{22} = 0 \tag{25}$$

Using the relation (20), the post-collision cumulants become:

$$\begin{aligned}
&C^*_{00} = C_{00} + \omega_{00}\left(C^{n\,eq}_{00} - C_{00}\right)\\
&C^*_{10} = C_{10} + \omega_{10}\left(C^{n\,eq}_{10} - C_{10}\right) + (1-\frac{\omega_{10}}{2})F_x\\
&C^*_{01} = C_{01} + \omega_{01}\left(C^{n\,eq}_{01} - C_{01}\right) + (1-\frac{\omega_{01}}{2})F_y\\
&C^*_{20} = C_{20} + \omega_{20}\left(C^{n\,eq}_{20} - C_{20}\right)\\
&C^*_{02} = C_{02} + \omega_{02}\left(C^{n\,eq}_{02} - C_{02}\right)\\
&C^*_{11} = C_{11} + \omega_{11}\left(C^{n\,eq}_{11} - C_{11}\right)\\
&C^*_{21} = C_{21} + \omega_{21}\left(C^{n\,eq}_{21} - C_{21}\right)\\
&C^*_{12} = C_{12} + \omega_{12}\left(C^{n\,eq}_{12} - C_{12}\right)\\
&C^*_{22} = C_{22} + \omega_{22}\left(C^{n\,eq}_{22} - C_{22}\right)
\end{aligned} \tag{26}$$

where $F_x = 0.0$, $F_y = g\beta\rho(T-T_m)$, $T_m = (T_h+T_c)/2$. In the present D2Q9 lattice model, after several trial and error steps the relaxation rates are set as: $\omega_{00} = 0, \omega_{10} = \omega_{01} = 1, \omega_{20} = \omega_{02} = \omega_{11} = 1/\tau, \omega_{21} = \omega_{12} = 0.6, \omega_{22} = 1.4$. Where $\tau = \frac{\nu}{C_s^2} + \frac{1}{2}$ is the relaxation time which is calculated from the fluid kinematic viscosity ν. Here $C_s = \frac{1}{\sqrt{3}}$ is the lattice speed for the D2Q9 lattice model.

Extracting post-collision central moments from the post-collision cumulants (26) as:

$$\begin{aligned}
&K^*_{00} = \exp(C^*_{00}) = 1, \quad K^*_{10} = 0, \quad K^*_{01} = 0,\\
&K^*_{20} = C^*_{20}, \quad K^*_{02} = C^*_{02}, \quad K^*_{11} = C^*_{11},\\
&K^*_{21} = C^*_{21}, \quad K^*_{12} = C^*_{12}\\
&K^*_{22} = C^*_{22} + C^*_{20}C^*_{02} + 2C^*_{11}C^*_{11}
\end{aligned} \tag{27}$$

Transferring back the post-collision central moments (27) to the post-collision raw-moments as follows:

$$\begin{aligned}
&\mu^+_{00} = K^*_{00}\\
&\mu^+_{10} = K^*_{10} + u\\
&\mu^+_{01} = K^*_{01} + v\\
&\mu^+_{20} = K^*_{20} + u^2\\
&\mu^+_{02} = K^*_{02} + v^2\\
&\mu^+_{11} = K^*_{02} + uv\\
&\mu^+_{21} = K^*_{21} + vK^*_{20} + 2uK^*_{11} + u^2v\\
&\mu^+_{12} = K^*_{12} + uK^*_{02} + 2vK^*_{11} + uv^2\\
&\mu^+_{22} = K^*_{22} + 2vK^*_{21} + 2uK^*_{12} + v^2K^*_{20} + u^2K^*_{02} + 4uvK_{11} + u^2v^2
\end{aligned} \tag{28}$$

Re-scaling the post-collision raw-moments (28) multiply by the density distribution ρ as:

$$\begin{aligned}
&\mu^*_{00} = \rho\mu^+_{00}, \quad \mu^*_{10} = \rho\mu^+_{10}, \quad \mu^*_{01} = \rho\mu^+_{01}\\
&\mu^*_{20} = \rho\mu^+_{20}, \quad \mu^*_{02} = \rho\mu^+_{02}, \quad \mu^*_{11} = \rho\mu^+_{11}\\
&\mu^*_{21} = \rho\mu^+_{21}, \quad \mu^*_{12} = \rho\mu^+_{12}, \quad \mu^*_{22} = \rho\mu^+_{22}
\end{aligned} \tag{29}$$

The mapping between velocity and moment spaces can be transformed by linear transformations $\mu^* = Mf^*$ and $f^* = M^{-1}\mu^*$. The basis matrix M for D2Q9 lattice model is:

$$M = \left[|1\rangle, |q_\xi\rangle, |q_\eta\rangle, |q_\xi^2\rangle, |q_\eta^2\rangle, |q_\xi q_\eta\rangle, |q_\xi^2 q_\eta\rangle, |q_\xi q_\eta^2\rangle, |q_\xi^2 q_\eta^2\rangle\right]^T.$$

Finally, the post-collision distribution function f^* is calculated using the post-collision raw-moments (29) as follows:

$$
\begin{aligned}
f_0^* &= \left(\mu_{00}^* - \mu_{20}^* - \mu_{02}^* + \mu_{22}^*\right)\\
f_1^* &= \frac{1}{2}\left(\mu_{10}^* + \mu_{20}^* - \mu_{12}^* - \mu_{22}^*\right)\\
f_2^* &= \frac{1}{2}\left(\mu_{01}^* + \mu_{02}^* - \mu_{21}^* - \mu_{22}^*\right)\\
f_3^* &= \frac{1}{2}\left(-\mu_{10}^* + \mu_{20}^* + \mu_{12}^* - \mu_{22}^*\right)\\
f_4^* &= \frac{1}{2}\left(-\mu_{01}^* + \mu_{02}^* + \mu_{21}^* - \mu_{22}^*\right)\\
f_5^* &= \frac{1}{4}\left(\mu_{11}^* + \mu_{21}^* + \mu_{12}^* + \mu_{22}^*\right)\\
f_6^* &= \frac{1}{4}\left(-\mu_{11}^* + \mu_{21}^* - \mu_{12}^* + \mu_{22}^*\right)\\
f_7^* &= \frac{1}{4}\left(\mu_{11}^* - \mu_{21}^* - \mu_{12}^* + \mu_{22}^*\right)\\
f_8^* &= \frac{1}{4}\left(-\mu_{11}^* - \mu_{21}^* + \mu_{12}^* + \mu_{22}^*\right)
\end{aligned}
\tag{30}
$$

Now the steaming part has been implemented for shifting the momentum distribution functions move to the neighboring nodes as:

$$
f_i(\mathbf{r} + \mathbf{q_i}\Delta t, t + \Delta t) = f_i^*(\mathbf{r}, t) \tag{31}
$$

After applying appropriate boundary conditions on the distribution functions the updated macroscopic quantities the fluid density and velocities, and the pressure field are calculated as follows:

$$
\rho = \sum_i f_i, \quad \rho\mathbf{u} = \sum_i f_i \mathbf{q_i} + \frac{\mathbf{F}}{2} \tag{32}
$$

4.2. D2Q5 thermal cumulant lattices Boltzmann method

The discretized LB equation for the temperature field is written as:

$$
g_i(\mathbf{r} + \mathbf{q_i}\Delta t, t + \Delta t) - g_i(\mathbf{r}, t) = \Omega_i^* \tag{33}
$$

The D2Q5 lattice model for the passive scalar, the central moments can be calculated as follows:

$$
\begin{aligned}
&K_{00} = T = \sum_i g_i, \quad K_{10} = \sum_i (\xi_i - u) g_i, \quad K_{01} = \sum_i (\eta_i - v) g_i,\\
&K_{20} = \sum_i (\xi_i - u)^2 g_i, \quad K_{02} = \sum_i (\eta_i - v)^2 g_i
\end{aligned}
\tag{34}
$$

The cumulants for the D2Q5 model for temperature as:

$$
\begin{aligned}
&C_{00} = k_{00}, \quad C_{10} = K_{10}, \quad C_{01} = K_{01}\\
&C_{20} = K_{20} - K_{10}K_{10}, \quad C_{02} = K_{02} - K_{01}K_{01}
\end{aligned}
\tag{35}
$$

The cumulant equilibrium distribution functions are:

$$
C_{00}^{eq} = T, C_{10}^{eq} = C_{01}^{eq} = 0, C_{20}^{eq} = C_{02}^{eq} = TC_T^2 \tag{36}
$$

where $C_T = \frac{1}{\sqrt{5}}$ is used for the D2Q5 central moment-based LBM in Molla et al. [5]. Using the relation (20), the post-collision cumulants become:

$$
\begin{aligned}
C_{00}^* &= C_{00} + \omega_{00}\left(C_{00}^{eq} - C_{00}\right)\\
C_{10}^* &= C_{10} + \omega_{10}\left(C_{10}^{eq} - C_{10}\right)\\
C_{01}^* &= C_{01} + \omega_{01}\left(C_{01}^{eq} - C_{01}\right)\\
C_{20}^* &= C_{20} + \omega_{20}\left(C_{20}^{eq} - C_{20}\right)\\
C_{02}^* &= C_{02} + \omega_{02}\left(C_{02}^{eq} - C_{02}\right)
\end{aligned}
\tag{37}
$$

In the present D2Q5 thermal lattice model, the relaxation rates are set as: $\omega_{00} = 0, \omega_{10} = \omega_{01} = 1/\tau_\alpha, \omega_{20} = \omega_{02} = 1$. Where $\tau_\alpha = 5\alpha + \frac{1}{2}$ is the relaxation time which is calculated from the fluid thermal diffusivity $\alpha = \frac{Pr}{\nu}$.

Extracting post-collision central moments from the post-collision cumulants (37) as:

$$
\begin{aligned}
&K_{00}^* = C_{00}^*, \quad K_{10}^* = C_{10}^*, \quad K_{01}^* = C_{01}^*\\
&K_{20}^* = C_{20}^* + C_{10}^* C_{10}^*, \quad K_{02}^* = C_{02}^* + C_{01}^* C_{01}^*
\end{aligned}
\tag{38}
$$

Transferring back the post-collision central moments (38) to the post-collision raw-moments as follows:

$$
\begin{aligned}
\mu_{00}^+ &= K_{00}^*\\
\mu_{10}^+ &= K_{10}^* + K_{00}^* u\\
\mu_{01}^+ &= K_{01}^* + K_{00}^* v\\
\mu_{20}^+ &= K_{20}^* - K_{00}^* u^2 + 2K_{10}^* v + 2K_{00}^* uv\\
\mu_{02}^+ &= K_{02}^* - K_{00}^* v^2 - 2K_{01}^* u - 2K_{00}^* uv
\end{aligned}
\tag{39}
$$

Finally, the post-collision distribution function g^* is calculated using the post-collision raw-moments (38) as follows:

$$
\begin{aligned}
g_0^* &= \left(\mu_{00}^+ - \mu_{20}^+ - \mu_{02}^+\right)\\
g_1^* &= \frac{1}{2}\left(\mu_{10}^+ + \mu_{20}^+\right)\\
g_2^* &= \frac{1}{2}\left(\mu_{01}^+ + \mu_{02}^+\right)\\
g_3^* &= \frac{1}{2}\left(-\mu_{10}^+ + \mu_{20}^+\right)\\
g_4^* &= \frac{1}{2}\left(-\mu_{01}^+ + \mu_{02}^+\right)
\end{aligned}
\tag{40}
$$

Now the steaming part has been implemented for shifting the momentum distribution functions move to the neighboring nodes as:

$$
g_i(\mathbf{r} + \mathbf{q_i}\Delta t, t + \Delta t) = g_i^*(\mathbf{r}, t) \tag{41}
$$

4.3. Boundary conditions

For the no-slip wall, the bounce-back boundary condition is applied to the velocity fields as follows:

East:

$$
f_3 = f_1, f_7 = f_5, f_6 = f_8 \tag{42}
$$

West:

$$
f_1 = f_3, f_5 = f_7, f_8 = f_6 \tag{43}
$$

North:

$$
f_4 = f_2, f_7 = f_5, f_8 = f_6 \tag{44}
$$

South:

$$
f_2 = f_4, f_5 = f_7, f_6 = f_8 \tag{45}
$$

At a wall with a fixed temperature (T_w), the following boundary condition is used for the D2Q5 model cumulative thermal lattice Boltzmann simulation:

$$
g_i(\mathbf{r}, t + \Delta t) = -g_i^c(\mathbf{r}, t) + C_T^2 T_w \tag{46}
$$

For an adiabatic wall, the anti-bounce-back condition is applied:

$$
g_i(\mathbf{r}, t + \Delta t) = g_i^c(\mathbf{r}, t) \tag{47}
$$

4.4. Convergence criteria

In this CuLBM, the iterative procedure is stopped when the velocity and temperature fields meet the following convergence criteria:

$$
\frac{\sum |\Phi^{(k+1)} - \Phi^{(k)}|}{\sum |\Phi^{(k+1)}|} < 10^{-12} \tag{48}
$$

where (Φ) represents the velocity ($\mathbf{u}$) or temperature (T), (k) is the iteration index, and the sum is taken over the entire domain.

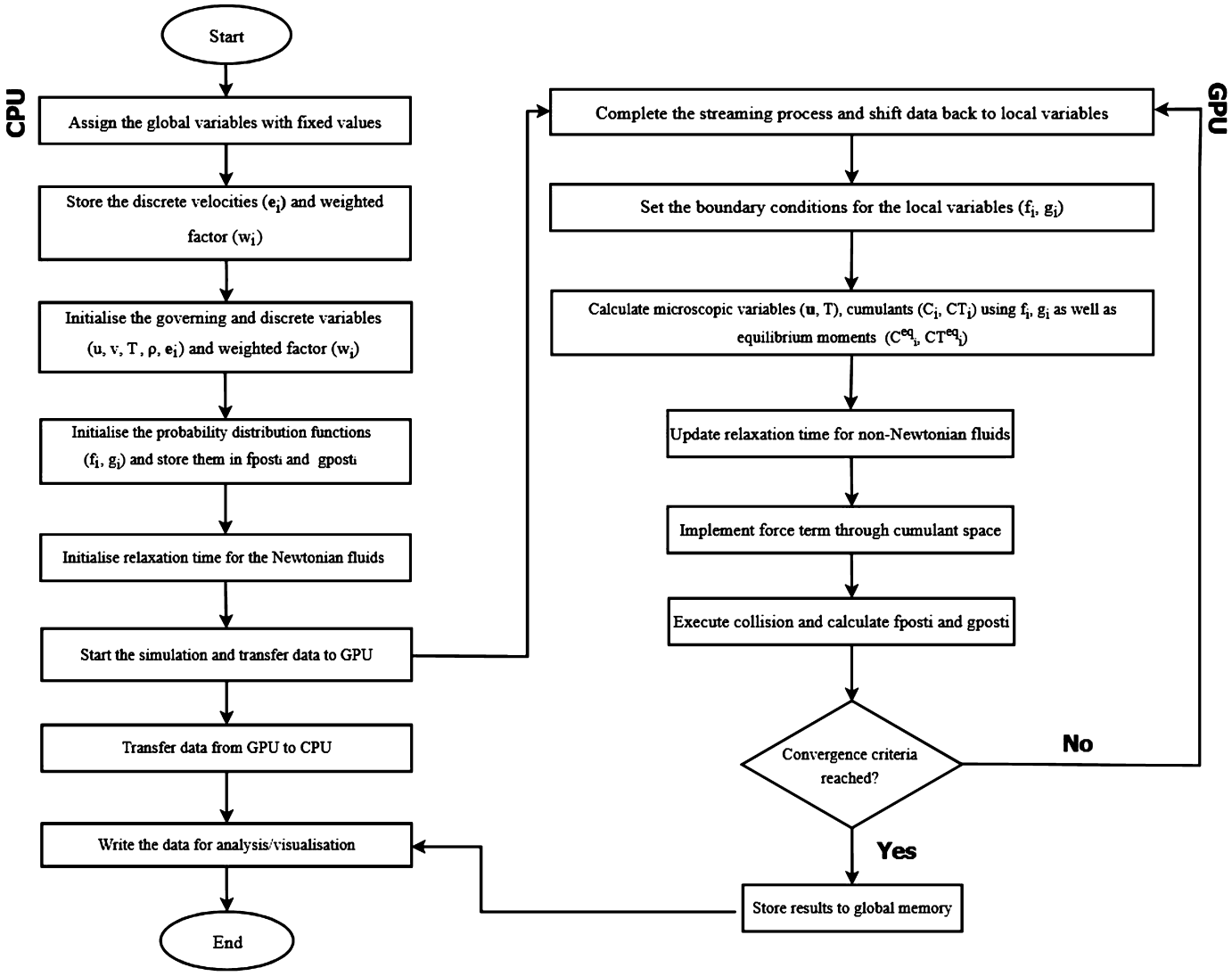


Fig. 2. A flowchart for the GPU based cumulant LBM.

After applying boundary conditions on the distribution functions the updated macroscopic quantities the fluid temperature is calculated as follows:

$$T = \sum_i g_i \tag{49}$$

4.5. Local and average rate of heat transfer

In the heat transfer problem, we are interested in calculating the rate of heat transfer using the local and average Nusselt numbers, which are defined below:

$$Nu(y) = -\frac{\partial \theta}{\partial x}\Big|_w \tag{50}$$

and

$$\overline{Nu} = \frac{1}{H}\int_0^H Nu(y)dy \tag{51}$$

In the current simulation, two different NVIDIA GPUs were utilized: the Quadro RTX 6000 with 24 GB of RAM and the GeForce RTX 4080 SUPER with 16 GB of RAM. The CPU used is an Intel(R) Core(TM) i7-9700K running at 3.60 GHz, accompanied by 128 GB of RAM. In a previous study by Molla et al. [31], a detailed description of the GPU implementation of the MRT-LBM was provided. A flowchart for the GPU based CuLBM is given here in Fig. 2. The CUDA C++-based two-dimensional cumulant lattice Boltzmann method (CuLBM) has been validated for the side heated square cavity in the subsequent subsections:

5. Numerical parameters and stability considerations

(i) The computational domain was divided into a uniform grid with a resolution of $N \times N$. The value of N was carefully selected based on grid independence studies to strike a balance between accuracy and computational efficiency. In this simulation, we set $\Delta x = \Delta y = 1$.

(ii) The time step in Lattice Boltzmann Method (LBM) is closely tied to both the lattice spacing and the relaxation time. For our simulations, we opted for a time step of $\Delta t = 1$, which aligns with the typical practices in LBM.

(iii) The relaxation parameters, particularly for the cumulant-based model, were chosen with great care to ensure numerical stability while also reducing numerical diffusion.

(iv) To keep the scheme stable, the relaxation time based the kinematics and thermal diffusivity τ, has been kept above 0.5, and the maximum Mach number, Ma, well under 0.1.

Table 1
Grid independence test for $Ra = 10^7$ while $Pr = 0.71$.

Lattice size:	(128 × 128)	(192 × 192)	(256 × 256)	(384 × 384)	(512 × 512)
$\overline{Nu}$	15.7731	16.3208	16.4836	16.5082	16.4981
u_{max}	152.1568	152.1204	150.9232	149.9438	149.0132
y	0.8740	0.8691	0.8745	0.8746	0.8786
v_{max}	693.3334	696.1868	699.2823	697.6656	698.2347
x	0.01574	0.02094	0.01961	0.0188	0.0196
$\lvert\psi_{max}\rvert$	30.0129	30.0883	30.1048	30.1262	30.1361
x	0.9063	0.9115	0.9102	0.9115	0.9121
y	0.4531	0.4427	0.4453	0.4453	0.4453

Table 2
Grid independence test for $Ra = 10^8$ while $Pr = 0.71$.

Lattice size:	(384 × 384)	(512 × 512)	(640 × 640)	(768 × 768)	(896 × 896)
$\overline{Nu}$	30.0408	30.1611	30.2556	30.3261	30.4007
u_{max}	373.7587	363.7762	349.1837	343.9222	343.6898
y	0.9441	0.9413	0.9374	0.9361	0.9341
v_{max}	2219.5027	2218.4246	2222.3586	2224.4225	2227.7573
x	0.0112	0.0117	0.0109	0.0117	0.0111
$\lvert\psi_{max}\rvert$	53.8235	53.8684	53.8848	53.9259	53.9915
x	0.0468	0.0468	0.0468	0.0481	0.0479
y	0.5535	0.5546	0.5546	0.5534	0.5535

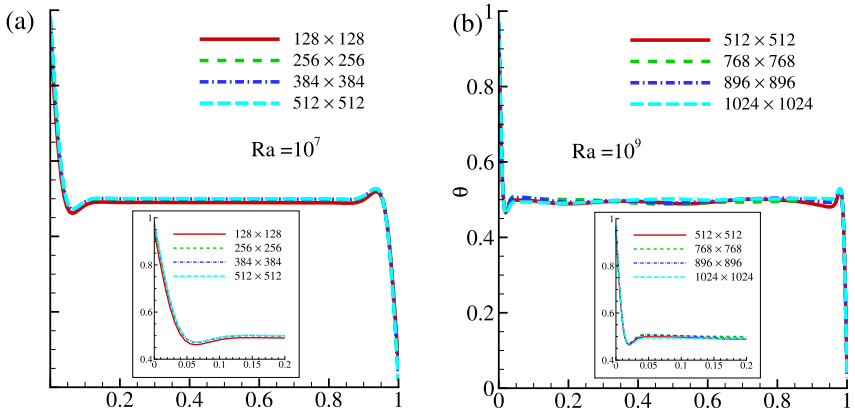


Fig. 3. Grid independence test in terms of the temperature distribution (a) $Re = 10^7$ and (b) $Re = 10^9$ while $Pr = 0.71$ at $y/H = 0.5$.

(v) It has also been performed grid convergence tests to verify the independence of results from the mesh resolution, and residuals were monitored to confirm the attainment of steady-state solutions while $Ra \leq 10^8$ and time-averaged solutions for $Ra = 10^9$.

6. Grid independence test

The grid independence test for $Ra = 10^7$ and $Ra = 10^8$ at $Pr = 0.71$, as described in Tables 1 and 2, focus on the impact on the flow parameters - average Nusselt number ($\overline{Nu}$), maximum velocities (u_{max}, v_{max}), their location on space (x, y), and the maximum stream function ($|\psi_{max}|$) – as a result of changes in lattice resolution. These tests guarantee numerical precision as well as independence to the grid for the cumulant LBM simulations. In computational fluid dynamics, an essential grid dependency test should be conducted before further analysis. First, one grid independence test is performed for the Rayleigh numbers $Ra = 10^7$ by using five different lattice sets as shown in Table 1. The results show that these parameters tend to stabilize with finer resolution, indicating numerical convergence. The finer grids demonstrate only small differences, indicating that the chosen resolution is sufficient to capture the flow dynamics. The tabulated results for the different physical quantities also show that the lattice size of (256×256) is sufficient for $Ra \leq 10^7$.

A similar grid independence test was done for $Ra = 10^8$ in Table 2. As the Rayleigh number increases, so does the strength of the convective effects, and sharper gradients are produced. It is necessary to have an adequately fine grid, or else the flow structures will not be resolved. The change of $\overline{Nu}$, u_{max}, v_{max} and their spatial coordinates with $|\psi_{max}|$ was analyzed at different resolutions. The results show that at a specific grid resolution, more refinement does not significantly change the parameters, confirming grid independence. The chosen resolution (640×640) is the best trade-off between computation time and accuracy for $Ra = 10^8$.

Here a further test is conducted for the two separate Rayleigh numbers $Ra = 10^7$ and $Ra = 10^9$, whose results in terms of temperature distribution are illustrated in Fig. 3(a)-(b). For $Ra = 10^7$, four different grid sets $(128 \times 128, 256 \times 256, 384 \times 384, 512 \times 512)$ were tested and it was found that 256×256 is adequate for $Ra \leq 10^7$. Another grid dependency test was performed for $Ra = 10^9$, where $512 \times 512, 768 \times 768, 896 \times 896, 1024 \times 1024$ were the grid sizes, and this test confirmed that 768×768 is appropriate for $Ra = 10^9$. These tests show the confidence in the cumulant LBM for capturing natural convection phenomena at various Ra, where the grid resolutions

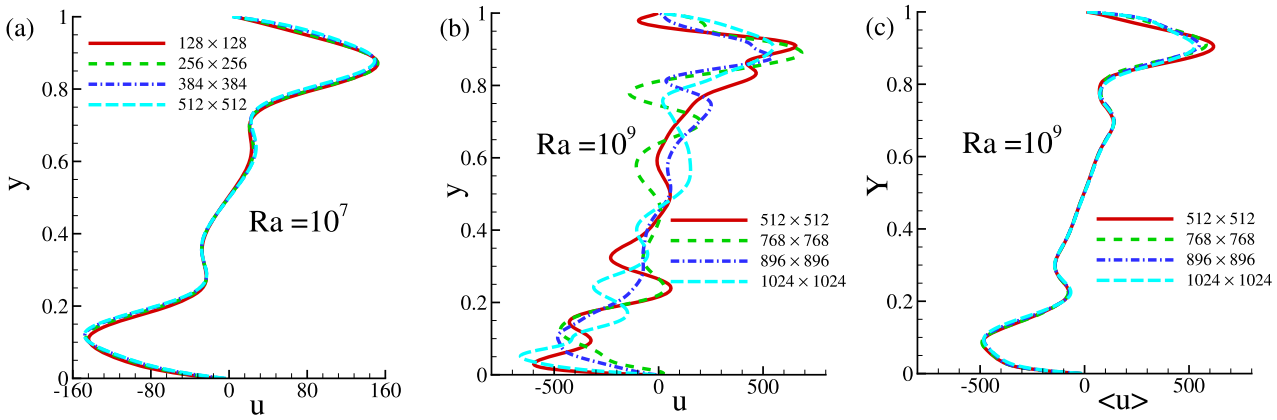


Fig. 4. Grid independence test in terms of the u-velocity distribution (a) steady-state results for $Re = 10^7$ and (b) instantaneous results for $Re = 10^9$, (c) time-averaged results for $Re = 10^9$ while $Pr = 0.71$ at $x/H = 0.5$.

were optimized to provide reasonable accuracy at minimal computational expense. In Fig. 4(a)-(c), another grid sensitivity analysis was done that looks at how grid resolution impacts the u-velocity distribution at the mid-section ($x/H = 0.5$) across various Rayleigh numbers (Ra) and Reynolds numbers (Re). We keep the Prandtl number constant at $Pr = 0.71$ for all scenarios. Let's break down each sub-figure. The u-velocity profiles are almost indistinguishable, showing that even the coarser grid (128×128) effectively captures the key flow characteristics when $Ra = 10^7$. However, for high-$Ra = 10^9$ flows, which fall into the turbulent regime, the solution becomes quite sensitive to grid resolution, making it impossible to achieve grid independence. In turbulent flows, it's crucial to test the time-averaged results for grid sensitivity. The time-averaged u-velocity profiles are significantly smoother and align much better across all grid resolutions compared to the instantaneous profiles shown in (b). Time-averaging effectively dampens high-frequency fluctuations, resulting in outcomes that are less affected by grid resolution. The profiles from the 768×768 grid and finer resolutions demonstrate convergence, suggesting grid independence for statistical quantities.

7. Benchmarking for method validation

Before dive into comparing the current results with what's already out there, it is obvious to normalize the present CuLBM results. This involves applying the right transformations: for velocity, it has been uses $\frac{\nu}{PrH}$; for temperature, it's $\frac{T-T_c}{T_h-T_c}$; and the stream function ψ is normalized by the thermal diffusivity α.

7.1. $10^3 \leq Ra \leq 10^6$

The validation of the current cumulant lattice Boltzmann method (LBM) is illustrated through a thorough comparison with established numerical benchmarks. These include the finite difference method (FDM) results from De Vahl Davis [18], the single-relaxation-time LBM (SRT-LBM) solutions by Dixit and Babu [20], and the multiple-relaxation-time LBM (MRT-LBM) data from Liu et al. [21], specifically for Rayleigh numbers ranging from 10^3 to 10^6 at $Pr = 0.71$, as shown in Table 3. The parameters evaluated—maximum horizontal velocity (u_{max}), maximum vertical velocity (v_{max}), their spatial coordinates (x, y), and the average Nusselt number ($\overline{Nu}$)—exhibit strong agreement across all methods. For example, at $Ra = 10^5$, the current u_{max} of 34.5884 is closely aligned with the FDM value of 34.730 and the MRT-LBM value of 34.6891. Additionally, the $\overline{Nu}$ of 8.8672 at $Ra = 10^6$ compares favorably with the FDM result of 8.800 and the MRT-LBM result of 8.8258. Minor discrepancies, such as the slightly higher v_{max} of 220.155 (current) compared to 219.362 (FDM) at $Ra = 10^6$, likely arise from variations in grid resolution (with 256×256 for $Ra = 10^5$) or numerical schemes. The consistency of results across all Ra cases, especially the positional coordinates (x, y) of velocity maxima, highlights the robustness and accuracy of the cumulant LBM in simulating natural convection. This validation affirms the method's reliability within the tested Ra range and its ability to replicate benchmark solutions with high precision.

7.2. $10^7 \leq Ra \leq 10^9$

The validation of the current cumulant lattice Boltzmann method (CuLBM) for high Rayleigh numbers ($10^7 \leq Ra \leq 10^9$) at $Pr = 0.71$ is shown through comparisons with the spectral difference method (SDM) by Le Quéré [19] and the single-relaxation-time LBM (SRT-LBM) of Dixit and Babu [20], as detailed in Table 4. Key parameters—maximum horizontal and vertical velocities (u_{max}, v_{max}), their spatial coordinates (x, y), and the average Nusselt number ($\overline{Nu}$)—were assessed across increasingly refined grids (up to 1024×1024 for $Ra = 10^9$).

For u_{max}, the current results at $Ra = 10^7$ (159.3020) and $Ra = 10^8$ (364.8981) are in close agreement with SDM (148.58, 321.876) and SRT-LBM (164.236, 389.877), effectively bridging the gap between the two methods. At $Ra = 10^9$, the current $u_{max} \approx 547.9$ (for 1024×1024 grid) surpasses SRT-LBM's 503.24, indicating a better resolution of velocity extremes at ultrahigh Ra. The y-coordinates of u_{max} (e.g., 0.8666 at $Ra = 10^7$) closely correspond with SDM (0.879) and SRT-LBM (0.851), while at $Ra = 10^9$, results stabilize around 0.89–0.91, reflecting trends in grid convergence.

For v_{max}, there is excellent agreement at $Ra = 10^7$ (current: 699.282 vs. SDM: 699.236) and $Ra = 10^8$ (2218.706 vs. SDM: 2222.39). At $Ra = 10^9$, the current method predicts $v_{max} \approx 7194.93$, which exceeds SRT-LBM's 6820.07, likely.

The results for $\overline{Nu}$ at $Ra = 10^7$ (16.4836) and $Ra = 10^8$ (30.1664) align closely with those from SRT-LBM (16.790, 30.506) but show a notable divergence from SDM (16.523, 30.225). This small discrepancy could be attributed to variations in the methods used for computing $\overline{Nu}$ or how boundary conditions are handled. At $Ra = 10^9$, the current $\overline{Nu} \approx 54.1$ is a bit lower than SRT-LBM's 57.35, which might be influenced by differences in turbulence modeling or grid resolution. The stability of $\overline{Nu}$ across finer grids (54.06-54.58) suggests strong convergence.

This validation highlights the cumulant LBM's effectiveness in simulating high-Ra flows, capturing finer structures as grid refinement occurs and extending predictions to $Ra = 10^9$, where benchmark data is limited. The minor discrepancies observed reflect inherent methodological differences, while the overall consistency reinforces the reliability of this method for studying turbulent natural convection. Here, it should be mentioned that the multiple-relaxation-time for the momentum LB equation is determined by a kinematic viscosity $\nu = 0.02$ for $10^3 \leq Ra \leq 10^7$ and $\nu = 0.005$ for $Ra \geq 10^8$.

Table 3
Comparison of the present results with FDM solution by De Vahl Davis [18], SRT-LBM solution of Dixit and Babu [20] and MRT-LBM results of Liu et al. [21] while $10^3 \leq Ra \leq 10^6$ and $Pr = 0.71$.

Ra		10^3	10^4	10^5	10^6
Grid used		(128×128)	(128×128)	(256×256)	(256×256)
u_{max}	FDM [18]	3.469	16.178	34.730	64.630
	SRT [20]	3.6529	16.163	35.521	64.186
	MRT [21]	3.6528	16.1770	34.6891	64.6968
	Present	3.6533	16.1463	34.5884	64.7212
y	FDM [18]	0.813	0.823	0.855	0.850
	SRT [20]	0.8125	0.823	0.8554	0.8496
	MRT [21]	0.8125	0.8229	0.8555	0.8516
	Present	0.8156	0.8237	0.8549	0.8509
v_{max}	FDM [18]	3.697	19.617	68.590	219.36
	SRT [20]	3.682	19.569	68.655	219.866
	MRT [21]	3.7004	19.6184	68.5034	219.149
	Present	3.7144	19.5863	68.5871	220.155
x	FDM [18]	0.178	0.119	0.066	0.0379
	SRT [20]	0.1718	0.125	0.0664	0.0371
	MRT [21]	0.1797	0.1198	0.0664	0.0391
	Present	0.1764	0.1178	0.0665	0.0372
$\overline{Nu}$	FDM [18]	1.118	2.243	4.519	8.800
	SRT [20]	1.121	2.286	4.5463	8.652
	MRT [21]	1.1161	2.2447	4.5273	8.8258
	Present	1.1215	2.2551	4.5449	8.8672

Table 4
Comparison of the present results with spectral difference method (SDM) solution by [19] and SRT-LBM solution of Dixit and Babu [20] while $10^7 \leq Ra \leq 10^9$ and $Pr = 0.71$.

Ra		10^7	10^8	10^9	10^9	10^9	10^9
Grid used		(256×256)	(512×512)	(512×512)	(768×768)	(896×896)	(1024×1024)
u_{max}	SDM [19]	148.58	321.876	–	–	–	–
	SRT [20]	164.236	389.877	503.24	–	–	–
	Present	159.3020	364.8981	639.6830	544.4773	550.1056	547.910
y	SDM [19]	0.879	0.928	–	–	–	
	SRT [20]	0.851	0.937	0.966	–	–	–
	Present	0.8666	0.9412	0.9158	0.8905	0.9061	0.8963
v_{max}	SDM [19]	699.236	2222.39	–	–	–	–
	SRT [20]	701.922	2241.374	6820.07	–	–	–
	Present	699.282	2218.706	7194.931	7032.988	7074.764	7069.9256
x	SDM [19]	0.021	0.012	–	–	–	–
	SRT [20]	0.020	0.0112	0.0064	–	–	–
	Present	0.019	0.0117	0.0058	0.0065	0.0067	0.0059
$\overline{Nu}$	SDM [19]	16.523	30.225	–	–	–	–
	SRT [20]	16.790	30.506	57.3505	–	–	–
$\overline{Nu}$	SDM [32]	–	–	–	–	54.6	–
	Present	16.4836	30.1664	54.0775	54.2237	54.5871	54.0629

7.3. Error analysis for the average Nusselt number

The error analysis shown in Table 5 provides a solid quantitative check of the current CuLBM (Cumulant Lattice Boltzmann Method) results against benchmark solutions found in the literature. Specifically, it compares the results from the FDM (Finite Difference Method) by De Vahl Davis [18] and the CTLBM results by Sharma et al. [22] for the average Nusselt number, Nu¯Nu, at various Rayleigh numbers (Ra), while keeping the Prandtl number fixed at $Pr = 0.71$.

For all Rayleigh numbers examined, the relative percentage errors between the CuLBM results and the reference values are impressively low, usually staying below 1%. This indicates a strong agreement between the CuLBM and established numerical methods, reinforcing the accuracy and reliability of the solver we've developed. When we compare the CuLBM against both FDM (De Vahl Davis [18]) and CTLBM (Sharma et al. [22]), its consistent error performance highlights its robustness across different numerical methods.

The comparisons with Sharma et al. [22] (at higher Ra values ranging from 10^7 to 10^9) still reveal relative errors under 0.5%, which further supports its reliability in high-Ra scenarios. This analysis bolsters the validation of the CuLBM code, suggesting that the method effectively captures the characteristics of natural convection in square cavities, even across a broad spectrum of Ra values. It demonstrates that CuLBM can replicate benchmark solutions with minimal deviation, which is crucial for the credibility of CFD, especially in simulations involving thermally driven flows.

Table 5
Error calculation among the results of FDM solution by De Vahl Davis [18], CTLBM solution of Sharma et al. [22] and the present CuLBM for the average Nusselt number ($\overline{Nu}$) while $Pr = 0.71$.

	Average Nusselt number ($\overline{Nu}$)		
Ra	De Vahl Davis [18]	CuLBM	|Error|%
10^3	1.118	1.1215	0.313
10^4	2.243	2.2551	0.539
10^5	4.5463	4.519	0.600
10^6	8.800	8.8672	0.764
Ra	Sharma et al. [22]	CuLBM	|Error|%
10^7	16.52722	16.4836	0.264
10^8	30.31308	30.1664	0.484
10^9	54.8217	54.5871	0.428

8. Results and discussion

The current findings regarding the streamlines, isotherms, velocity and temperature distributions, as well as the local and average Nusselt numbers and entropy generation for both Newtonian and non-Newtonian fluids in the flowing subsections are presented here.

8.1. Streamlines and isotherms for the Newtonian fluid

A side-heated square cavity serves as a classic benchmark problem in natural convection, where the temperature difference between the vertical walls drives fluid motion. Typically, the left wall is hot while the right wall is cold, resulting in buoyancy-driven convection currents. The flow structures and thermal gradients are significantly influenced by the Rayleigh number (Ra), which indicates the relative strength of buoyancy forces compared to viscous damping. Fig. 5 (a)-(f) illustrates the streamlines and isotherms for $Ra = 10^4$ to $Ra = 10^6$. In Fig. 5 (a), at $Ra = 10^4$, the flow remains steady, showcasing a single dominant circulation cell. The streamlines reveal a primary vortex centered in the cavity, with weaker secondary recirculation zones near the corners. The flow is relatively smooth, with gradual transitions between areas of different velocities. In Fig. 5(b), the isotherms appear mostly vertical in the central region, indicating that conduction is the dominant mode of heat transfer. Near the walls, the isotherms bend slightly due to convective transport, but the temperature gradient remains significant, suggesting a moderate convective effect. As we move to $Ra = 10^5$, the flow strengthens, and the convection effect becomes more pronounced. The primary vortex transforms into an elongated structure, and secondary vortices emerge near the corners. The circulation intensifies, resulting in higher velocity gradients and increased mixing, as seen in Fig. 5(c). In Fig. 5(d), the isotherms become more curved, particularly near the vertical boundaries, indicating enhanced convective transport. The temperature gradient near the walls decreases slightly as convection takes precedence over pure conduction. In the core region, some conduction effects are still noticeable. At $Ra = 10^6$, the flow is very vigorous, characterized by strong convective motion. The main vortex becomes more elongated, and additional smaller vortices emerge due to heightened flow instabilities. The boundary layers close to the walls are thinner, indicating higher velocity gradients. The isotherms exhibit significant distortions near the hot and cold walls, highlighting the increased influence of convection. A thin thermal boundary layer develops near the walls, while the central area of the cavity maintains a relatively uniform temperature due to effective mixing. At lower Ra, conduction prevails, and streamlines create a simple, symmetric vortex structure. As Ra increases, convection intensifies, resulting in more intricate flow patterns with multiple vortices (as shown in Fig. 5(e)-(f)). The isotherms shift from a nearly vertical configuration (indicating conduction dominance) to a more distorted shape, reflecting stronger convective mixing. The boundary layers become thinner as Ra rises, suggesting enhanced heat transfer rates.

As the Rayleigh number exceeds 10^6, buoyancy-driven convection becomes increasingly dominant, resulting in the formation of complex flow structures and thinner thermal boundary layers, as shown in Fig. 6(a)-(f). This change signifies a transition from laminar to turbulent convection, characterized by more intricate vortical structures and sharper temperature gradients. In Fig. 6(a), at $Ra = 10^7$, the flow displays vigorous convective motion, with multiple vortices developing near the walls. The main circulation cell is significantly deformed, and smaller eddies can be seen forming in the corners. The streamlines appear more irregular, suggesting the beginning of unsteady flow behavior. In Fig. 6(b), the isotherms are closely packed near the vertical walls, indicating strong thermal gradients. The central region is dominated by convective mixing, leading to nearly horizontal temperature contours. The boundary layers at both the hot and cold walls become thinner, which is typical of increased heat transfer rates. In Figs. 6(c)-(d), at $Ra = 10^8$, the flow shows considerable turbulence. The primary vortex is significantly distorted, and many small vortices are present throughout the cavity, particularly in the upper and lower sections. The flow near the boundaries is highly sheared, suggesting the existence of thin viscous layers. The isotherms exhibit even sharper gradients near the hot and cold walls, further decreasing the thickness of the thermal boundary layer. The temperature contours in the core region become more wavy and irregular, reflecting enhanced convective mixing and turbulent fluctuations. At $Ra = 10^9$, the flow transitions to a fully turbulent state, characterized by chaotic structures and unsteady eddies throughout the cavity. The boundary layers become extremely thin, and significant fluctuations in the velocity field are observed. Various recirculation zones develop in different areas, which further promote mixing. The isotherms display highly irregular patterns, reflecting strong convective disturbances and turbulent heat transfer. While the boundary layers remain very thin, the central region experiences minimal temperature variation due to the intense mixing, as illustrated in Figs. 6(e)-(f). As Ra increases, the flow shifts from a steady laminar state to a turbulent regime, with a growing number of vortices and unsteady structures. The thermal boundary layers along the vertical walls become progressively thinner as buoyancy-driven convection intensifies. Heat transfer increasingly relies on convection rather than conduction, evident in the deformation and waviness of the isotherms.

8.2. Velocity and temperature distribution

The vertical velocity distribution illustrated in Fig. 7(a)-(d) shows how the velocity component v/U varies along the x-axis at a mid-plane height ($y/H = 0.5$). The Rayleigh number (Ra) plays a crucial role in controlling buoyancy-driven convection, which has a significant impact on the velocity profile. As Ra increases, the flow transitions from being dominated by conduction to being dominated by convection. At lower Ra, conduction prevails, resulting in smooth velocity profiles. However, as Ra rises, convection intensifies, leading to notable velocity variations, steeper gradients, and the potential for turbulence. At $Ra = 10^3$, the velocity profile remains relatively smooth and symmetric, indicating a steady, laminar convection process. As Ra increases to 10^4 and 10^5, the velocity peak around $x \approx 0.1$ becomes more pronounced. By the time we reach $Ra = 10^6$, oscillations start to appear, signaling the onset of unsteady convective motion. Although the peak velocity increases with Ra, the variations are still moderate compared to even higher Ra values. For lower Ra values (10^3 to 10^5), conduction remains dominant, and the effects of convection are weak. At $Ra = 10^6$, a more robust convective current develops, resulting in larger velocity variations and local accelerations near the boundaries. In Fig. 7(c)-(d), at $Ra = 10^7$, the peak vertical velocity rises sharply. For $Ra = 10^8$, the velocity fluctuations become steeper, and the peak shifts towards smaller x. By $Ra = 10^9$, turbulence-like behavior starts to emerge, characterized by strong velocity gradients and high peak magnitudes. The shift

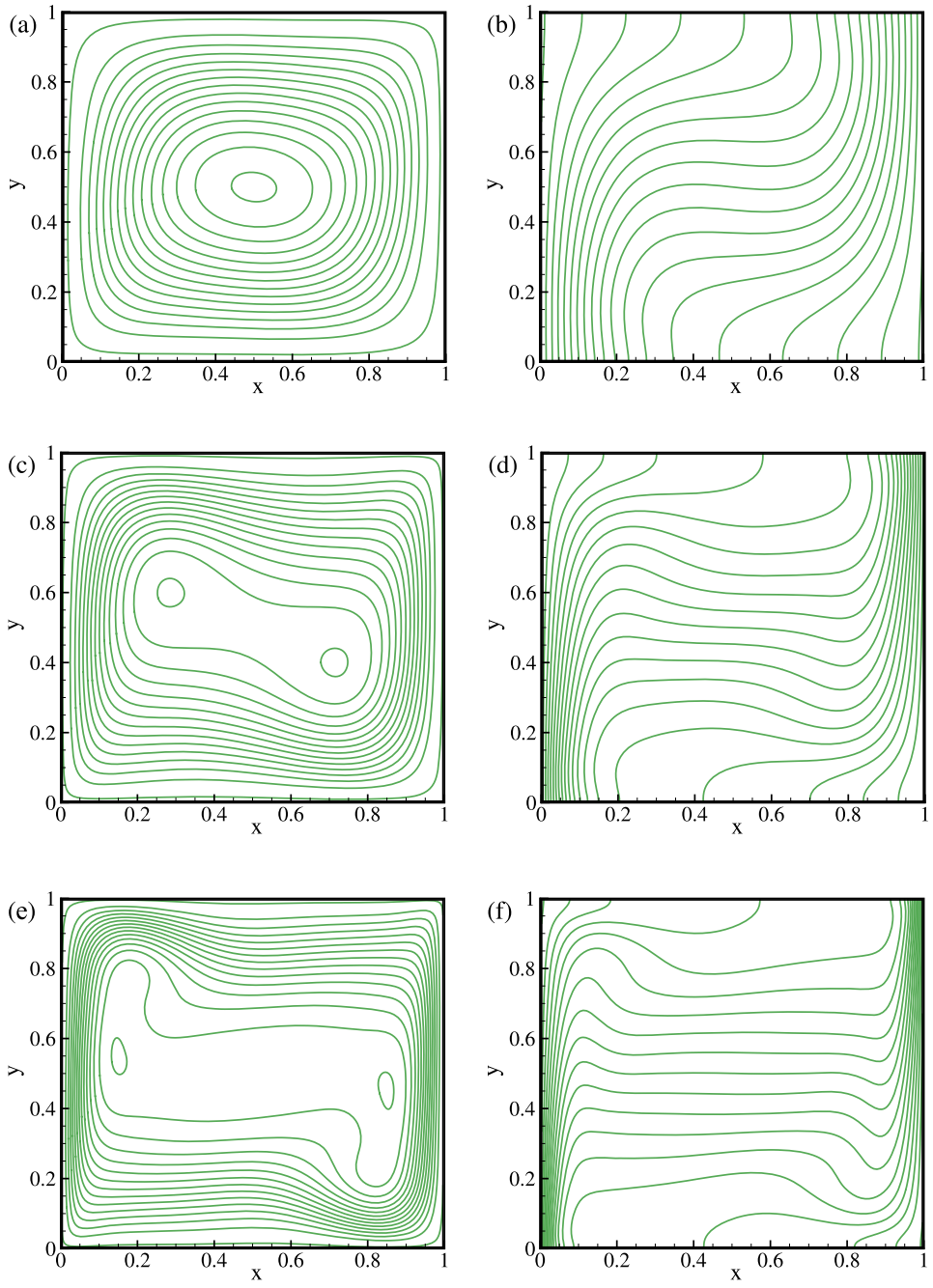


Fig. 5. Streamlines (left) isotherms (right): (a) and (b) for $Re = 10^4$, (c) and (d) for $Re = 10^5$, (e) and (f) for $Re = 10^6$.

from smooth to steep gradients indicates an increase in flow instability. At $Ra = 10^7$, a strong convective cell forms, although the flow retains some structure. As we move to $Ra = 10^8$ and $Ra = 10^9$, turbulence begins to take over, resulting in highly localized velocity peaks and sharp fluctuations. The leftward shift of the velocity peak suggests a growing influence of boundary layer effects. Laminar regime where $Ra < 10^6$, the flow is stable, exhibiting a clear velocity profile. Transitional regime where $Ra \sim 10^6$ to 10^7, small oscillations and secondary flow structures emerge, signaling the beginning of turbulence. Turbulent regime while $Ra > 10^8$, significant velocity fluctuations and sharp gradients indicate fully developed turbulent convection.

The temperature distribution (θ) at various Rayleigh numbers ($Ra = 10^3$ to $Ra = 10^9$) along the x-axis at mid y ($y/H = 0.5$) is illustrated in Fig. 8(a)-(d). The Rayleigh number plays a crucial role in determining the shift from conduction-dominated to convection-dominated heat transfer. At lower Ra, conduction prevails, resulting in a linear temperature profile. As Ra increases, the influence of convection intensifies, leading to the formation of boundary layers and temperature stratification. In Fig. 8(a) and (b), for $Ra = 10^3$, the temperature profile remains almost linear, signifying a conduction-dominated state. When Ra rises to 10^4 and 10^5, noticeable deviations from linearity emerge, with steeper temperature gradients observed near the boundaries. At $Ra = 10^6$, the temperature gradient near the right boundary becomes more pronounced, indicating the emergence of convective thermal plumes. A sharp temperature drop is noted near $x \approx 0.8$, reflecting enhanced convective transport. In Fig. 8(c) and (d), for $Ra = 10^7$, a significant temperature gradient develops near the left boundary. At $Ra = 10^8$, the gradient steepens further, and the temperature in the central region approaches uniformity. By the time we reach $Ra = 10^9$, the thermal boundary layer is extremely thin, with temperature transitions occurring

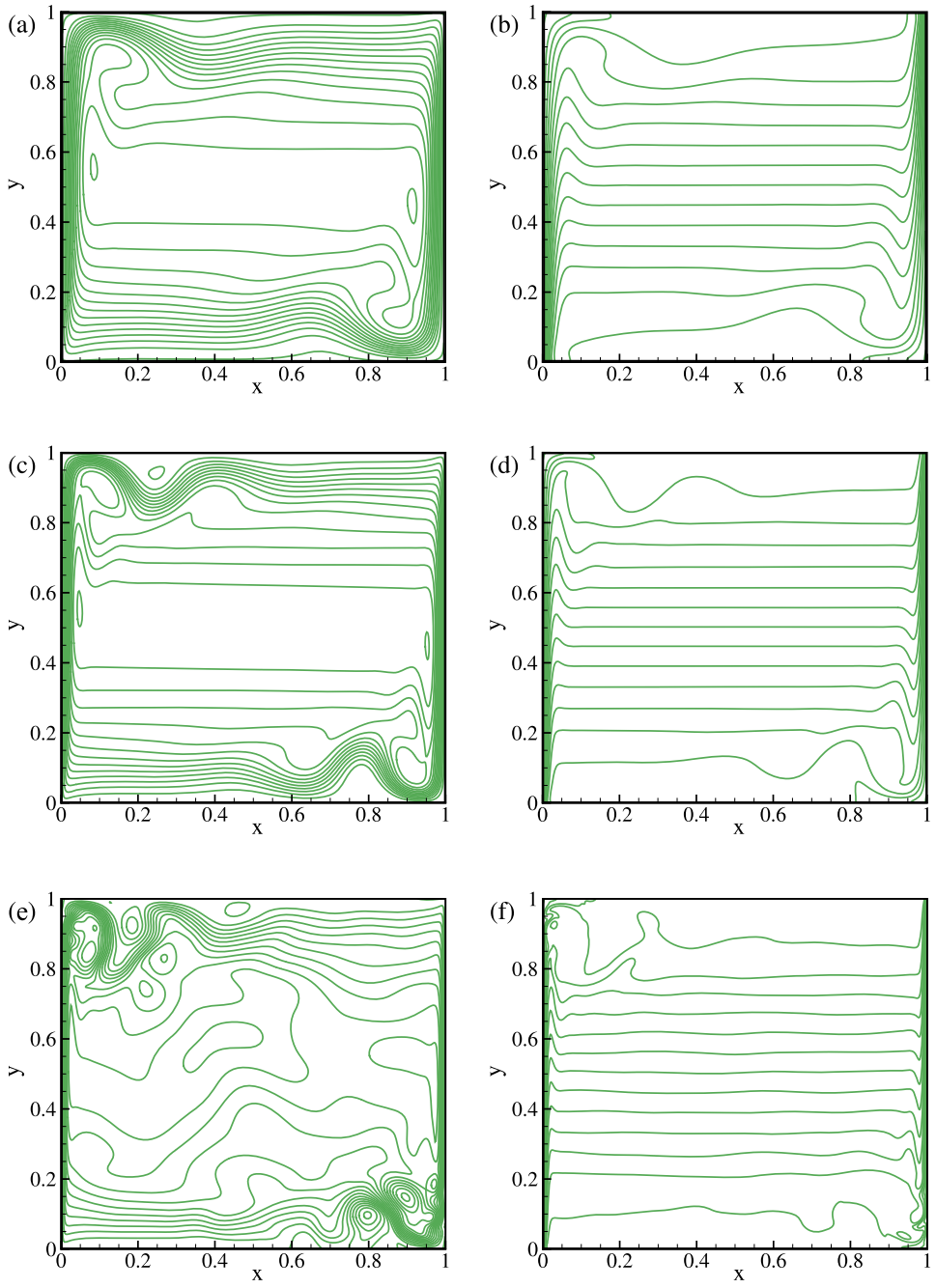


Fig. 6. Streamlines (left) isotherms (right): (a) and (b) for $Re = 10^7$, (c) and (d) for $Re = 10^8$, (e) and (f) for $Re = 10^9$.

over a very short distance. The temperature field exhibits stratification, highlighting a clear distinction between hot and cold areas. At elevated Ra, convection takes over, resulting in pronounced stratification. The thermal boundary layers thin out, while the core region remains nearly isothermal, with a sharp temperature drop occurring near the left wall, indicative of strong convective transport.

8.3. Local Nusselt number

Figs. 9(a)-(b) illustrate how the local Nusselt number (Nu) changes along the heated left wall of a square cavity for various Rayleigh numbers ($Ra = 10^3$ to $Ra = 10^9$). The local Nusselt number (Nu) indicates the rate of heat transfer due to convection, with higher Nu values signifying stronger convective heat transfer, while lower values suggest that conduction is the primary mode of heat transfer. In Fig. 9(a), for $Ra = 10^3$ (solid red line), the local Nu remains low along the wall, indicating that conduction is the dominant mechanism. At $Ra = 10^4$ (green dashed line), Nu shows a slight increase near the bottom, hinting at the beginning of convection. For $Ra = 10^5$ and $Ra = 10^6$ (blue and pink lines), a significant convective heat transfer effect is observed, resulting in much higher Nu values near the bottom. The Nu decreases along the height: the maximum heat transfer occurs at the bottom of the cavity, tapering off towards the top as the boundary layer develops and thermal gradients diminish. In Fig. 9(b), for $Ra = 10^7$ (solid red) and $Ra = 10^8$ (green dashed), the stronger convection effects yield higher Nu values at the bottom. At $Ra = 10^9$ (blue dash-dot line), intense convection leads to a peak at the bottom, with fluctuations near the top suggesting instabilities or secondary flow structures. The Nu values are significantly elevated: at very high Ra, the flow transitions to turbulence, and convective motion becomes the primary driver of heat transfer. The transition from conduction to convection occurs as Ra in-

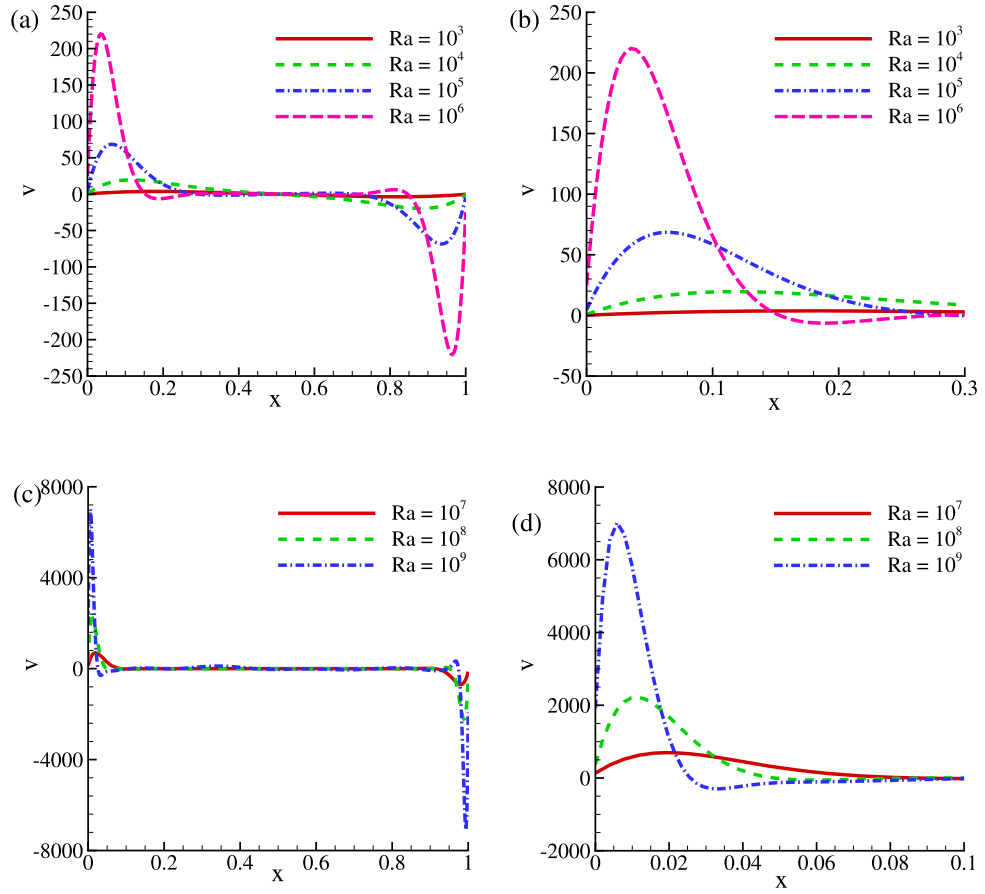


Fig. 7. Vertical velocity distribution for: (a) $Ra = 10^3$ to $Ra = 10^6$ for whole x-line, (b) $Ra = 10^3$ to $Ra = 10^6$ for x-line up to 0.3, (c) $Ra = 10^4$ to $Ra = 10^9$ for whole x-line, and (d) $Ra = 10^4$ to $Ra = 10^9$ for whole x-line up to 0.1 while at $y/H = 0.5$.

creases, with heat transfer initially dominated by conduction at low Ra, and buoyancy-driven convection taking over as Ra rises. Additionally, for higher Ra, a thin thermal boundary layer develops at the left wall, resulting in sharp variations in Nu near the bottom. Figs. 9(a)-(b) illustrate how the local Nusselt number (Nu) changes along the heated left wall of a square cavity for various Rayleigh numbers ($Ra = 10^3$ to $Ra = 10^9$). The local Nusselt number (Nu) indicates the rate of heat transfer due to convection, with higher Nu values signifying stronger convective heat transfer, while lower values suggest that conduction is the primary mode of heat transfer. For $Ra = 10^3$ (solid red line), the local Nu remains low along the wall, indicating that conduction is the dominant mechanism. At $Ra = 10^4$ (green dashed line), Nu shows a slight increase near the bottom, hinting at the beginning of convection. For $Ra = 10^5$ and $Ra = 10^6$ (blue and pink lines), a significant convective heat transfer effect is observed, resulting in much higher Nu values near the bottom. The Nu decreases along the height: the maximum heat transfer occurs at the bottom of the cavity, tapering off towards the top as the boundary layer develops and thermal gradients diminish. For $Ra = 10^7$ (solid red) and $Ra = 10^8$ (green dashed), the stronger convection effects yield higher Nu values at the bottom. At $Ra = 10^9$ (blue dash-dot line), intense convection leads to a peak at the bottom, with fluctuations near the top suggesting instabilities or secondary flow structures. The Nu values are significantly elevated: at very high Ra, the flow transitions to turbulence, and convective motion becomes the primary driver of heat transfer. The transition from conduction to convection occurs as Ra increases, with heat transfer initially dominated by conduction at low Ra, and buoyancy-driven convection taking over as Ra rises. Additionally, for higher Ra, a thin thermal boundary layer develops at the left wall, resulting in sharp variations in Nu near the bottom.

8.4. Benchmarking for the flow irreversibility

The non-dimensional form of the flow irreversibility due to the fluid friction (σ_F) and temperature gradient (σ_T) as in Ilis et al. [33] by, respectively:

$$S_F = \lambda_1 \left[2\left(\frac{\partial u}{\partial x}\right)^2 + 2\left(\frac{\partial v}{\partial y}\right)^2 + \left(\frac{\partial v}{\partial x} + \frac{\partial u}{\partial y}\right)^2 \right] \tag{52}$$

$$S_T = \left[\left(\frac{\partial \theta}{\partial x}\right)^2 + \left(\frac{\partial \theta}{\partial y}\right)^2 \right] \tag{53}$$

The comparison of entropy generation between the current cumulant-based lattice Boltzmann method (CuLBM) and the findings of Ilis et al. [33] provides several important insights regarding both fluid friction and temperature gradient contributions at $Ra = 10^5$ and $Pr = 0.71$ in Figs. 10(a)-(d). The spatial distribution of viscous dissipation entropy shows a strong correlation between the two studies, with peak values concentrated in areas of high shear stress, such as near vertical walls and in the wake of convective plumes. The magnitude of entropy generation in the LBM results closely matches Ilis et al.'s data, although minor discrepancies in peak values are noted. These differences may arise from the CuLBM's enhanced capability to resolve smaller-scale vortical structures and minimize numerical diffusion, resulting in sharper gradients in velocity fields. Overall, the fluid friction entropy patterns affirm the LBM's accuracy in modeling viscous effects, especially in capturing boundary layer dynamics and flow separation. Regarding entropy generation due to temperature gradient (Fig. 10(a) and Fig. 10(d)), both studies predict maximum thermal irreversibility in areas with steep temperature gradients, such as thermal boundary layers and plume interfaces. The LBM results show slightly more localized peaks, likely due

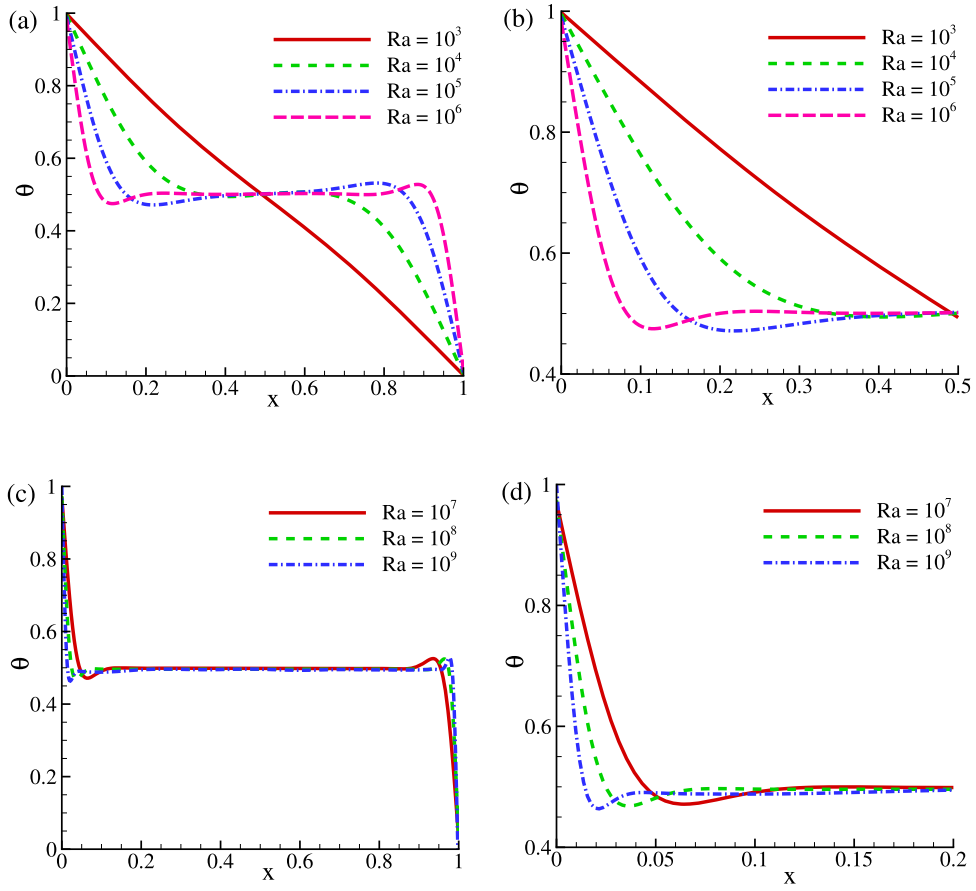


Fig. 8. Temperature distribution for: (a) $Ra = 10^3$ to $Ra = 10^6$ for whole x-line, (b) $Ra = 10^3$ to $Ra = 10^6$ for x-line up to 0.5, (c) $Ra = 10^4$ to $Ra = 10^9$ for whole x-line, and (d) $Ra = 10^4$ to $Ra = 10^9$ for whole x-line up to 0.2 while at $y/H = 0.5$.

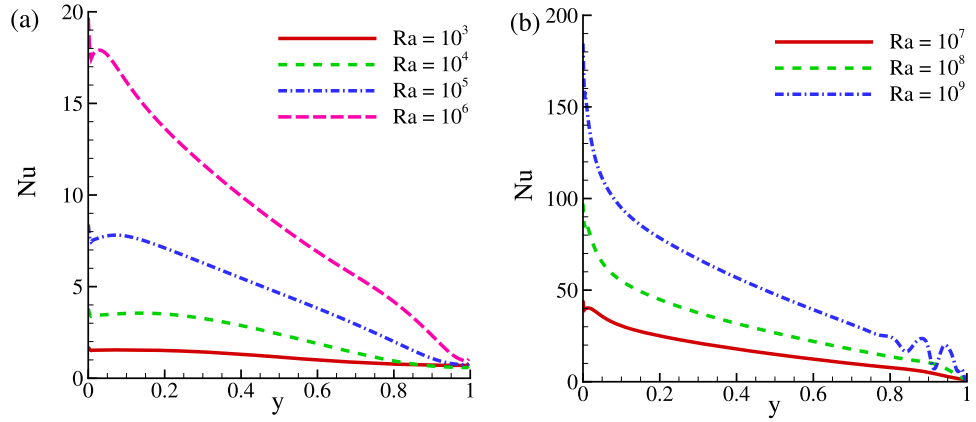


Fig. 9. Local Nusselt number for: (a) $Ra = 10^3$ to $Ra = 10^6$ and (b) $Ra = 10^4$ to $Ra = 10^9$.

to the method's inherent ability to resolve finer thermal structures with minimal numerical dissipation. The spatial agreement in thermal entropy distribution highlights the robustness of the LBM's thermal model, particularly in simulating conductive and convective heat transfer mechanisms. A notable consistency is observed in the relative dominance of temperature-driven entropy over fluid friction contributions, as reported by Ilis et al. [33], confirming that thermal irreversibility plays a key role in entropy generation at $Ra = 10^5$. The cumulant based LBM shows a strong correlation with the traditional numerical method used by Ilis et al. [33] for predicting entropy generation mechanisms. Although there are slight discrepancies in the peak values, the method's capability to capture intricate flow and thermal characteristics confirms its reliability for examining entropy generation in buoyancy-driven flows. This comparison underscores the LBM's potential as a valuable tool for investigating thermodynamic irreversibilities in fluid systems.

8.4.1. Efficiency gains of GPU over CPU:

Table 6 presents a comparative analysis of GPU-based parallel performance against sequential CPU performance, emphasizing the simulation time per iteration. The results demonstrate a substantial advantage of the GPU over the CPU, with speedup defined as the ratio of CPU to GPU simulation time. For example, at a lattice size of 64×64, the GPU achieves a speedup of approximately 11.07 times. This acceleration becomes even more pronounced at a lattice size of 2048×2048, reaching

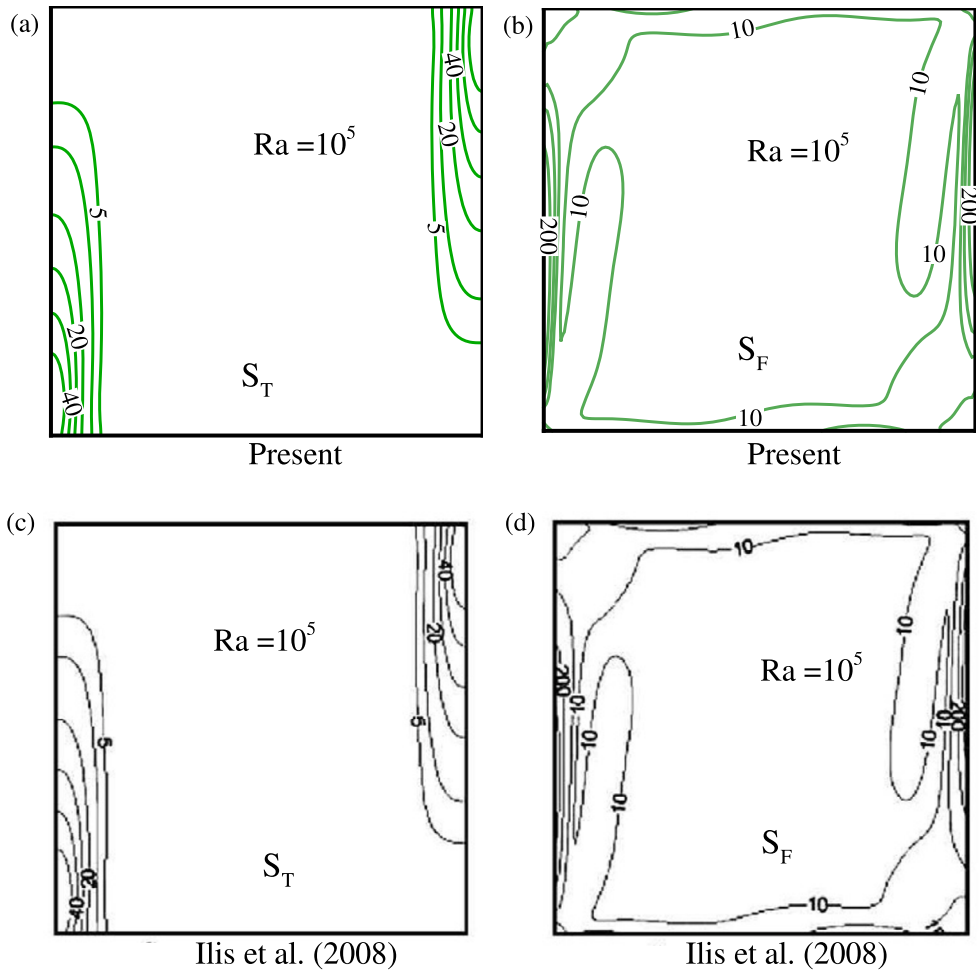


Fig. 10. Comparison of the entropy generation with the results of Ilis et al. [33] in terms of the fluid friction in (a) and (c), and due to the temperature gradient in (b) and (d) while $Ra = 10^5$, $\lambda_1 = 10^{-4}$ and $Pr = 0.71$.

Table 6
The CuLBM simulation times were recorded on both a CPU (Intel Core i7-9700K, 3.60 GHz, 128 GB RAM) and a GPU (RTX 4080 SUPER with 16 GB RAM and 10240 CUDA cores) for various lattice configurations of the side-heated cavity problem with Newtonian fluids, under conditions of $Ra = 10^6$ and $Pr = 0.71$ The performance comparison is based on 2000 simulation iterations.

Lattice size	Time (s): CPU	GPU	Speed up $= t_{CPU}/t_{GPU}$
64 ×64	0.970867	0.087655	11.07
128 ×128	4.824362	0.090379	53.38
256×256	21.186866	0.324198	65.35
512×512	249.784846	0.978909	255.17
1024×1024	1065.525134	3.8122	279.51
2048×2048	5168.042805	16.4292	314.56

a remarkable speedup of 314.59 times. These results underscore the strong dependence of GPU performance on grid size, with larger lattices yielding significantly greater speedup and computational efficiency than smaller ones.

9. Shear-rate dependent non-Newtonian fluids

To effectively simulate shear-rate dependent non-Newtonian fluids, we need to make some adjustments to the standard D2Q9 CuLBM for the velocity field. Drawing from our recent research on the CLBM [5], it's crucial to separate the bulk and shear viscosity terms in the central moments, and this separation also extends into the cumulant space. This distinction becomes particularly important when working with shear-thinning fluids, where the viscosity drops in comparison to Newtonian viscosity as shear rates increase.

To implement this decoupling, we modify the normalized central moment space by splitting the bulk and shear viscosity terms in equation (22). We need to make these adjustments to properly address the different ways we handle shear and bulk viscosities, which helps ensure that our model accurately captures the non-Newtonian flow characteristics found in shear-dependent fluids.

$$\begin{aligned}
&\bar{\hat{K}}_{00} = 1, \quad \bar{\hat{K}}_{10} = \hat{K}_{10}, \quad \bar{\hat{K}}_{01} = \hat{K}_{01} \\
&\bar{\hat{K}}_{20} = \hat{K}_{20} + \hat{K}_{02}, \quad \bar{\hat{K}}_{02} = \hat{K}_{20} - \hat{K}_{02}, \quad \bar{\hat{K}}_{11} = \hat{K}_{11} \\
&\bar{\hat{K}}_{21} = \hat{K}_{21}, \quad \bar{\hat{K}}_{12} = \hat{K}_{12}, \quad \bar{\hat{K}}_{22} = \hat{K}_{22}
\end{aligned} \tag{54}$$

It is possible to compute the updated accumulated values, denoted as $C_{\xi,\eta}$. The equilibrium cumulants have also undergone modifications, resulting in new values that reflect the adjustments in the system dynamics. These changes impact the equilibrium state, and further calculations may be necessary to determine their specific effects on the system.

$$C_{00}^{eq}=1, C_{10}^{eq}=C_{01}^{eq}=0, C_{20}^{eq}=2C_s^2, C_{02}^{eq}=0, C_{11}^{eq}=C_{21}^{eq}=C_{12}^{eq}=C_{22}^{eq}=0 \tag{55}$$

A modification is made to one of the relaxation parameters, ω_{20}, based on the bulk viscosity ν_b, which is defined as:

$$\nu_b=\left(\frac{1}{\omega_{20}}-0.5\right)C_s^2\Delta t \tag{56}$$

This approach follows the methodology employed by Suga et al. [34] in the D3Q27 MRT-LBM. After performing the post-collision calculations in cumulant space, we proceed by separating the bulk and shear viscosity components according to the method of Yahia and Premnath [35], as applied in the CLBM. This segregation process allows for more accurate modeling of the fluid's viscous behavior, where the distinct roles of bulk and shear viscosity are accounted for separately in the lattice dynamics.

$$\begin{aligned}\bar{C}_{20}^{*}&=\frac{1}{2}\left(C_{20}^{*}+C_{02}^{*}\right)\\ \bar{C}_{02}^{*}&=\frac{1}{2}\left(C_{20}^{*}-C_{02}^{*}\right)\end{aligned} \tag{57}$$

The remaining post-collision cumulants and overall method procedure remain unchanged. In the following subsections, we will validate the current CuLBM code by applying it to simulate two types of non-Newtonian fluids. These examples will help demonstrate the accuracy and effectiveness of the CuLBM approach in handling complex fluid behaviors beyond Newtonian assumptions.

The current CuLBM framework takes into account the unique flow characteristics of these fluids. Specifically:

(i) For power-law fluids, we've introduced a local viscosity model that updates the apparent viscosity at each lattice node, depending on the local shear rate and flow behavior index. This effectively captures the nuances of shear-thinning or shear-thickening behavior.

(ii) When it comes to Bingham fluids, we utilize a regularization method, like the Papanastasiou model, to approximate the yield stress behavior. This allows for a smooth transition between unyielded and yielded regions while ensuring numerical stability on the GPU.

These implementations will be integrated in the subsection 9.1 into the collision operator steps of the CuLBM algorithm, guaranteeing an accurate representation of non-Newtonian flow physics.

9.1. Non-Newtonian power-law (PL) fluids in a side heated square cavity

The governing dimensionless equations for the non-Newtonian power-law fluids (PL) are as follows [36,37]:

$$\frac{\partial u}{\partial x}+\frac{\partial v}{\partial y}=0 \tag{58}$$

$$\begin{aligned}&\frac{\partial u}{\partial \tau}+u\frac{\partial u}{\partial x}+v\frac{\partial u}{\partial y}=-\frac{\partial p}{\partial y}\\ &+\frac{Pr}{\left(\sqrt{Ra}\right)^{2-n}}\left\{\frac{\partial}{\partial x}\left(2\eta\frac{\partial u}{\partial x}\right)+\frac{\partial}{\partial y}\left(\eta\frac{\partial u}{\partial y}+\eta\frac{\partial v}{\partial x}\right)\right\}\end{aligned} \tag{59}$$

$$\begin{aligned}&\frac{\partial v}{\partial \tau}+u\frac{\partial v}{\partial x}+v\frac{\partial v}{\partial y}=-\frac{\partial p}{\partial y}\\ &+\frac{Pr}{\left(\sqrt{Ra}\right)^{2-n}}\left\{\frac{\partial}{\partial x}\left(\eta\frac{\partial v}{\partial x}+\eta\frac{\partial u}{\partial y}\right)+\frac{\partial}{\partial y}\left(2\eta\frac{\partial v}{\partial y}\right)\right\}+Pr\theta\end{aligned} \tag{60}$$

$$\frac{\partial \theta}{\partial \tau}+u\frac{\partial \theta}{\partial x}+v\frac{\partial \theta}{\partial y}=\frac{1}{\sqrt{Ra}}\left[\frac{\partial}{\partial x}\left(\frac{\partial \theta}{\partial x}\right)+\frac{\partial}{\partial y}\left(\frac{\partial \theta}{\partial y}\right)\right] \tag{61}$$

Here,

$$\eta=|\gamma|^{n-1}=\left\{2\left(\frac{\partial u}{\partial x}\right)^2+2\left(\frac{\partial u}{\partial y}\right)^2+\left(\frac{\partial u}{\partial y}+\frac{\partial v}{\partial x}\right)^2\right\}^{\frac{1}{2}(n-1)} \tag{62}$$

where the non-dimensional variables for the PL fluids are used to make the governing equations (1)-(4) dimensionless are as follows:

$$\begin{aligned}&x=\frac{\bar{x}}{H},\quad y=\frac{\bar{y}}{H},\quad u=\frac{\bar{u}}{\left(\frac{\alpha}{H}\right)\sqrt{Ra}},\quad v=\frac{\bar{v}}{\left(\frac{\alpha}{H}\right)\sqrt{Ra}}\\ &Ra=\frac{g\beta H^{2n+1}(T_h-T_c)}{\left(\frac{K}{\rho}\right)\alpha^n}\quad P=\frac{\bar{p}}{\rho\left(\frac{\alpha}{H}\right)^2 Ra},\quad \tau=\frac{\bar{t}\sqrt{Ra}}{\left(\frac{H^2}{\alpha}\right)}\\ &\theta=\frac{T-T_c}{T_h-T_c}\quad Pr=\frac{\left(\frac{K}{\rho}\right)}{\alpha^{2-n}H^{2(n-1)}}\end{aligned} \tag{63}$$

The PL fluid is a widely studied and practically significant class of non-Newtonian fluids with shear-dependent viscosity. In this class, viscosity varies as a function of shear rate and is described by the relation $\mu=K|\gamma|^{n-1}$, where γ represents the shear rate, n is the power-law index and K is the power-law consistency parameters, as detailed in previous studies [38–40]. For the shear-dependent non-Newtonian fluids, the power-law fluid is one of the importation complex fluids regarding the practical applications. Here we have considered the side-heated square cavity filled with the power-law fluids. The present CuLBM code is validated for the shear-thinning ($n<1$), Newtonian ($n=1$) and the shear-thickening ($n>1$) fluids. For the lattice Boltzmann method, the magnitude of the strain rate tensor $|\gamma|$ can be determined differently than the classical Navier-Stokes equations simulation. According to Aimon et al. [36], each node determines the shear rate or strain rate tensor γ_{ij} locally as follows:

$$\gamma_{ij}=-\frac{3}{2\rho c^2\tau_\nu(\mathbf{x},t)}\sum_{\alpha=0}^{8}\xi_{\alpha i}\eta_{\alpha j}\left(f_\alpha-f_\alpha^{eq}\right) \tag{64}$$

Here, f_α^{eq} is calculated from the equation (65)as in Mendu and Das [41].

$$f_i^{eq}=\rho w_i\left[1+\frac{(\mathbf{q}_i\cdot\mathbf{u})}{{C_s}^2}+\frac{(\mathbf{q}_i\cdot\mathbf{u})^2}{2{C_s}^4}-\frac{(\mathbf{u}\cdot\mathbf{u})}{2{C_s}^2}\right] \tag{65}$$

where w_i is the weighted factor for the D2Q9 model as in [41,36].

Figs. 11(a)-(b) offer a detailed comparison between the current CuLBM results and the numerical data shared by Turan et al. [38] regarding natural convection of non-Newtonian power-law fluids within a square cavity by considering the 256×256 lattice sizes. To validate these findings, it has been used two key benchmarks: the graph in Fig. 11(a) shows how the vertical velocity (v) changes along the horizontal line (x-axis) at the mid-height of the cavity ($y/H=0.5$) for a power-law index of $n=0.6$, a Prandtl number of $Pr=100$, and a Rayleigh number of $Ra=10^6$. The CuLBM results (depicted by the solid blue line) align remarkably well with the reference data from Turan et al. [38] (marked by red circles). This close match confirms the numerical accuracy and stability of our GPU-accelerated LBM solver. The velocity profile reveals pronounced boundary layers near the side walls, along with a central core that has relatively lower velocity gradients. The sharp velocity peaks close to the walls are captured with precision, showcasing CuLBM's ability to handle steep gradients in highly convective flows. Any minor discrepancies between the data-sets can likely be traced back to variations in spatial resolution or numerical discretization methods, but overall, the behavior remains consistent. Fig. 11(b) shows how the average Nusselt number ($\overline{Nu}$) changes along the hot wall based on the power-law index n for three different Rayleigh numbers: $Ra=10^4,10^5,10^6$, all while keeping $Pr=100$ constant. In this graph,

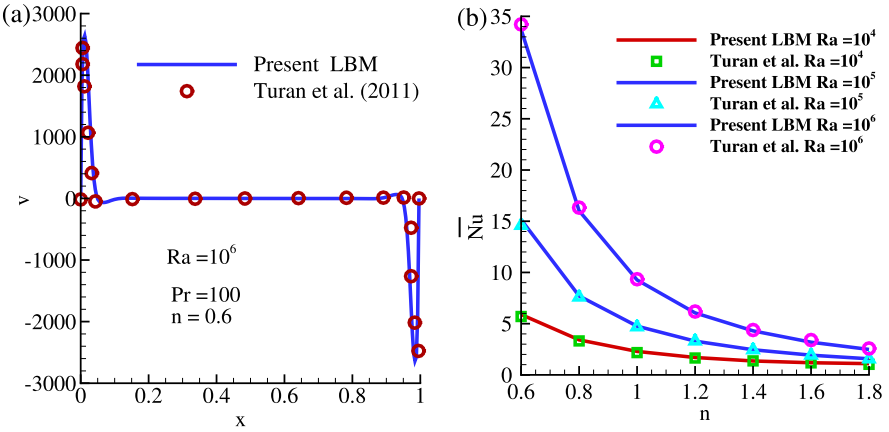


Fig. 11. Comparison of the present results with the results of finite volume method (FVM) by Turan et al. [38] (a) vertical velocity v while $n = 0.6$, $Pr = 100$ at $y/H = 0.5$ (b) average Nusselt number ($\overline{Nu}$) for the different $Ra = 10^4, 10^5, 10^6$ and the power-law index $n = 0.6, 0.8, 1.0, 1.2, 1.4, 1.6, 1.8$ while $Pr = 100$.

the results from the CuLBM simulations for $Ra = 10^4$, 10^5, and 10^6 are represented by red squares, green triangles, and magenta circles, respectively, and are compared to the findings from Turan et al. [38] (shown as blue lines). Across all Rayleigh numbers, we see that $\overline{Nu}$ decreases as the power-law index n increases. This pattern illustrates the shift from shear-thinning behavior ($n < 1$) to shear-thickening behavior ($n > 1$) in fluids. Shear-thinning fluids boost convective transport, which results in higher heat transfer rates, while shear-thickening fluids tend to hinder convective motion, leading to a drop in $\overline{Nu}$. The graph also clearly shows that as Ra increases, $\overline{Nu}$ rises, indicating that stronger buoyancy forces improve thermal transport. The findings from our CuLBM simulation closely match the reference values, confirming that our model effectively handles complex rheological behaviors with both accuracy and consistency. The strong agreement in both velocity and Nusselt number profiles highlights that the CuLBM approach is a solid and dependable method for simulating natural convection in non-Newtonian fluids.

Figs. 12(a)-(i) showcase how both the Rayleigh number ($Ra = 10^4, 10^5, 10^6$) and the power-law index ($n = 0.6, 1.0, 1.4$) impact the streamline patterns within a square cavity. These streamlines illustrate the flow structure created by buoyancy-driven convection of non-Newtonian fluids. All simulations are conducted at $Pr = 100$, ensuring that momentum diffusion takes precedence over thermal diffusion. In frames 12(a)-(c) where $Ra = 10^4$, the flow of shear-thinning fluids ($n < 1$) exhibits a notably stronger circulation compared to the Newtonian scenario. Here, we see two symmetric counter-rotating vortices, which are slightly distorted due to the lower viscosity near the walls. The strength of the streamlines (vorticity) is around |13.41|, indicating a moderate level of convection. In contrast, for Newtonian fluids ($n = 1$), a single symmetric central vortex forms, demonstrating weaker convection than the shear-thinning case. This flow is primarily influenced by diffusion, as shown by the lower streamline strength of approximately |5.14|. Meanwhile, shear-thickening fluids ($n > 1$) dampen flow circulation due to their increased effective viscosity. As a result, the streamlines become more concentric, and the central vortex weakens to about |2.25|, reflecting a decrease in buoyant motion. In frames 12(d)-(f) at $Ra = 10^5$, when $n = 0.6$, the flow starts to show a more asymmetric and intense pattern, particularly near the top and bottom walls. You can see two well-defined vortices forming, with a much stronger circulation of about |30.37|, which is influenced by the reduced viscosity from shear-thinning behavior. When $n = 1$, the central vortex becomes even stronger compared to the low Ra scenario, showing a streamline intensity of around |11.07|. The flow begins to display a greater degree of rotational symmetry, and for $n =$, it still maintains a single vortex structure, but it's a bit more centralized. The circulation improves compared to the $Ra = 10^4$ case, yet it still lags behind Newtonian and shear-thinning fluids due to the viscous resistance, which is approximately 5.75. In frames 12(g)-(i) for $Ra = 10^6$, when $n = 0.6$, we see that strong convection leads to the creation of thin boundary layers and elongated circulatory cells along the horizontal walls. The intensity of the streamlines is remarkably high (around |68.66|), indicating a vigorous flow driven by strong buoyancy forces and lower viscosity. For $n = 1.0$, in the Newtonian scenario, we observe a pair of symmetric corner eddies, with their vortex centers slightly shifted towards the side walls. The streamlines reveal secondary flow structures near the corners, showing a strength of about 19.57. However, when $n = 1.4$, the flow is the weakest of the three at this Rayleigh number. Here, a relatively weak and more diffusive central vortex emerges, with a maximum circulation intensity of around |2.95|. This finding highlights how shear-thickening behavior can limit the influence of thermal buoyancy on the development of flow.

Figs. 13(a)-(i) provide a visual representation of how isotherms, or temperature contours, are distributed, illustrating the heat transfer dynamics within a square cavity filled with non-Newtonian power-law fluids. As the Rayleigh number (Ra) increases from 10^4 to 10^6, we see the isotherms becoming increasingly distorted and compressed near the walls, particularly along the vertical boundaries. This shift suggests a move from heat transfer dominated by conduction to one driven more by convection. At lower Ra values (10^4), the isotherms remain mostly parallel and vertical, indicating that diffusive conduction is the primary mechanism at play. However, at higher Ra values (10^6), the isotherms start to bend significantly, revealing a strong influence of convective transport within the fluid. For shear-thinning fluids ($n = 0.6$), there's a noticeable horizontal layering of isotherms near the center, a result of enhanced convection, especially evident at high Ra. In contrast, Newtonian fluids ($n = 1.0$) display a transitional behavior that sits between conduction and convection. Meanwhile, shear-thickening fluids ($n = 1.4$) present more symmetric and less distorted isotherms, which suggests that the convective effects are weaker due to their higher viscosity at elevated shear rates.

The performance analysis presented in Table 7 compares two GPUs: GPU1, the Quadro RTX 6000 with 24 GB of RAM and 4608 CUDA cores, and GPU2, the GeForce RTX 4080 SUPER, which boasts 16 GB of RAM and a whopping 10240 CUDA cores. This analysis focuses on the side heated cavity problem using power-law fluids, with parameters set at $Ra = 10^6$, $Pr = 100$, and $n = 0.6$. The results show a speedup ranging from 1.31 to 1.46, clearly demonstrating that GPU2 consistently outshines GPU1 across all lattice sizes. Interestingly, the speedup tends to be slightly better for larger lattice sizes, which makes sense since GPU2's additional computational resources (like cores and memory bandwidth) are utilized more effectively as the problem size increases. With more than double the CUDA cores (10240 compared to 4608), GPU2 sig-

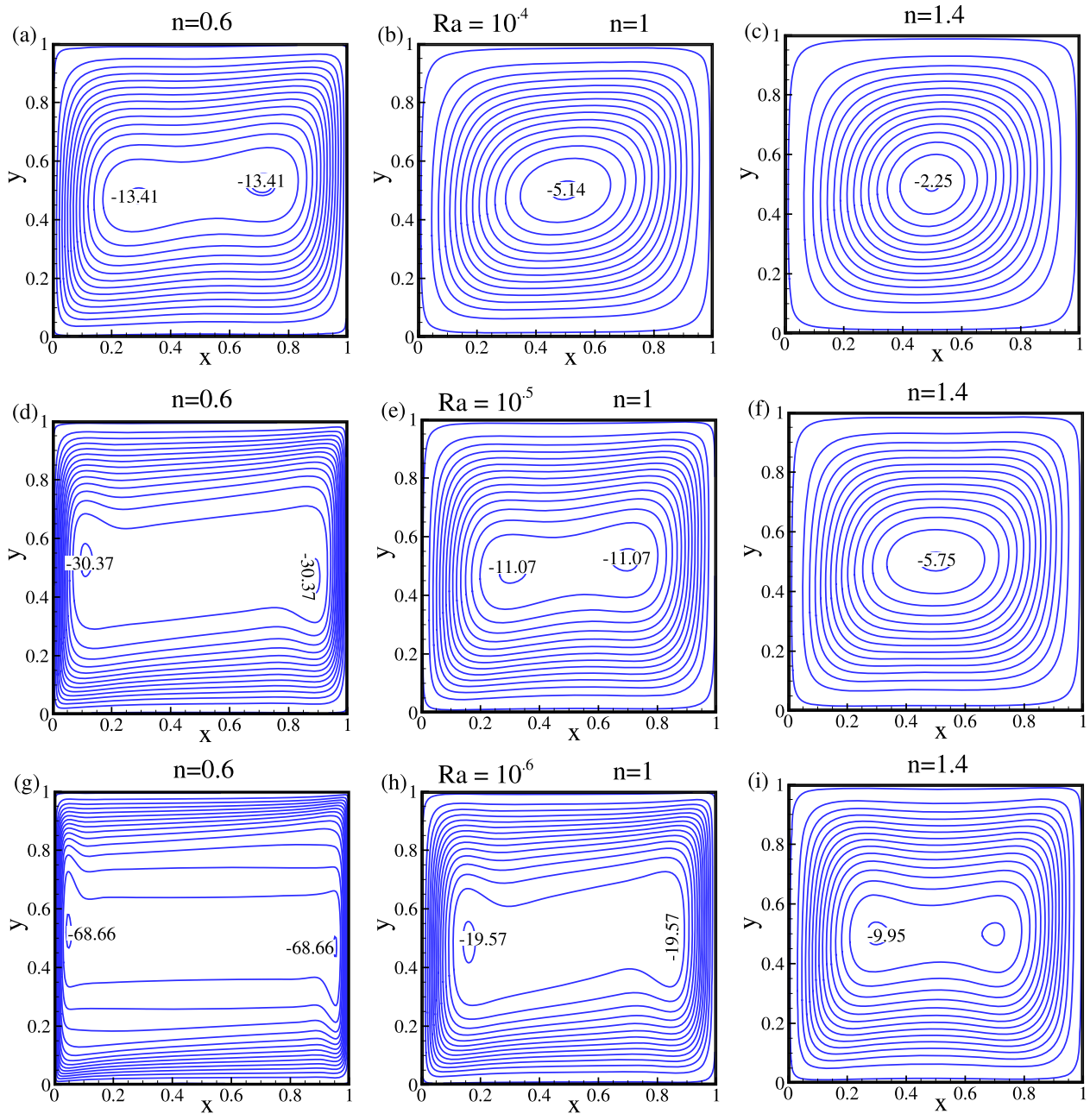


Fig. 12. Streamlines for the different Rayleigh numbers (Ra) and power-law index (n), (top) (a) $n = 0.6$ (b) $n = 1$ and (c) $n = 1.4$ for $Ra = 10^4$, (middle) (d) $n = 0.6$ (e) $n = 1$ and (f) $n = 1.4$ for $Ra = 10^5$ and (bottom) (g) $n = 0.6$ (h) $n = 1$ and (i) $n = 1.4$ for $Ra = 10^6$ while $Pr = 100$.

nificantly accelerates parallel computation. However, it's worth noting that the speedup isn't directly tied to the number of CUDA cores. Even though GPU1 has a larger memory capacity (24 GB), the faster memory (GDDR6X) and superior bandwidth of the 4080 SUPER seem to provide a greater advantage for this particular workload. The results indicate no memory bottlenecks, suggesting that the 16 GB on GPU2 is more than adequate for the largest domain tested (2048×2048).

9.2. Non-Newtonian Bingham fluids

The governing dimensionless equations for the non-Newtonian Bingham fluids are as follows [27]:

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0 \tag{66}$$

$$\frac{\partial u}{\partial \tau} + u\frac{\partial u}{\partial x} + v\frac{\partial u}{\partial y} = -\frac{\partial p}{\partial y} + \sqrt{\frac{Pr}{Ra}}\left\{\frac{\partial}{\partial x}\left(2\frac{\mu}{\mu_0}\frac{\partial u}{\partial x}\right) + \frac{\partial}{\partial y}\left(\frac{\mu}{\mu_0}\frac{\partial u}{\partial y} + \frac{\mu}{\mu_0}\frac{\partial v}{\partial x}\right)\right\} \tag{67}$$

$$\frac{\partial v}{\partial \tau} + u\frac{\partial v}{\partial x} + v\frac{\partial v}{\partial y} = -\frac{\partial p}{\partial y} + \sqrt{\frac{Pr}{Ra}}\left\{\frac{\partial}{\partial x}\left(\frac{\mu}{\mu_0}\frac{\partial v}{\partial x} + \frac{\mu}{\mu_0}\frac{\partial u}{\partial y}\right) + \frac{\partial}{\partial y}\left(2\frac{\mu}{\mu_0}\frac{\partial v}{\partial y}\right)\right\} + \theta \tag{68}$$

$$\frac{\partial \theta}{\partial \tau} + u\frac{\partial \theta}{\partial x} + v\frac{\partial \theta}{\partial y} = \frac{1}{\sqrt{PrRa}}\left[\frac{\partial}{\partial x}\left(\frac{\partial \theta}{\partial x}\right) + \frac{\partial}{\partial y}\left(\frac{\partial \theta}{\partial y}\right)\right] \tag{69}$$

where the non-dimensional variables for the PL fluids are used to make the governing equations (1)-(4) dimensionless are as follows:

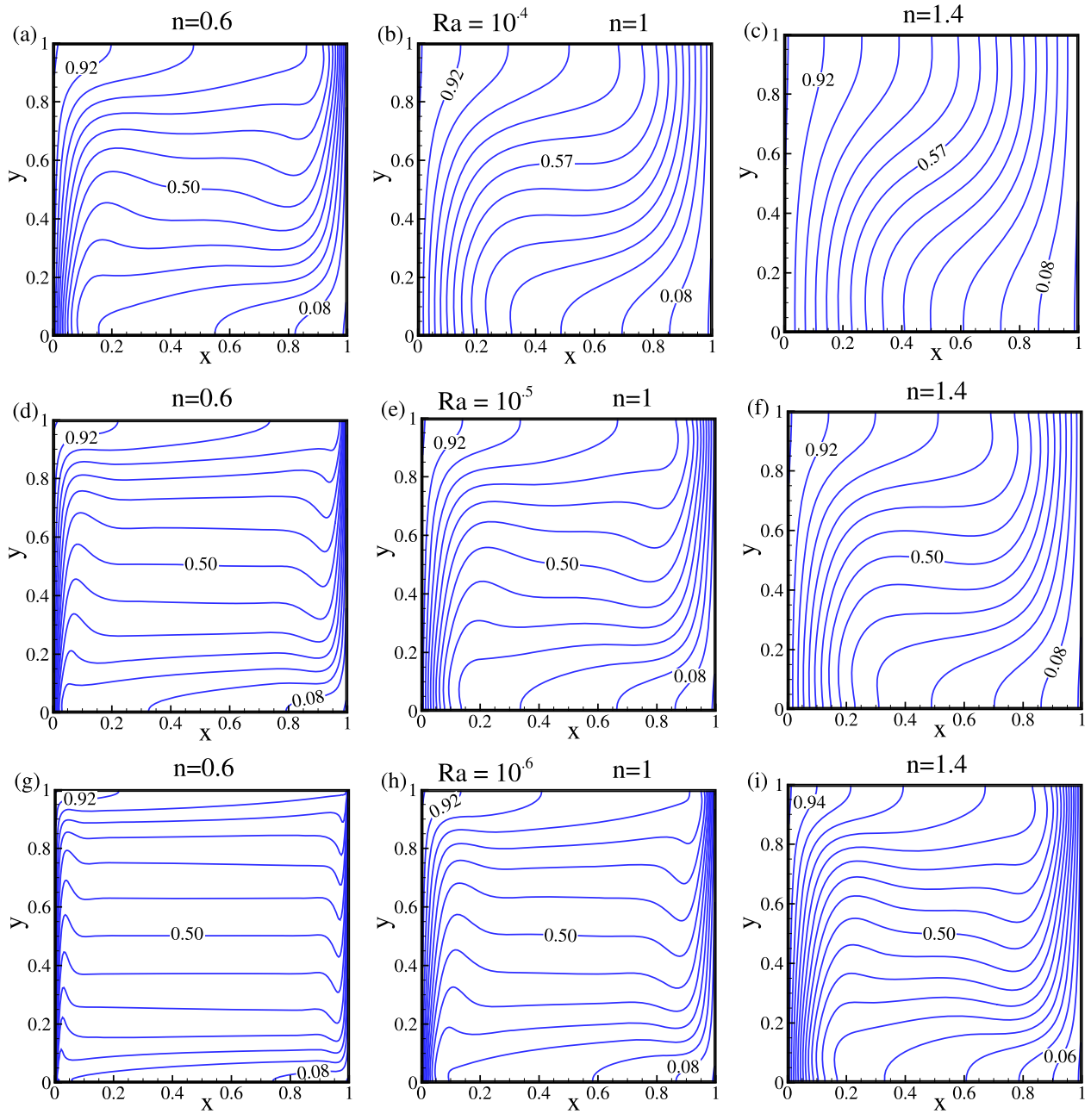


Fig. 13. Isotherms for the different Rayleigh numbers (Ra) and power-law index (n), (top) (a) $n = 0.6$ (b) $n = 1$ and (c) $n = 1.4$ for $Ra = 10^4$, (middle) (d) $n = 0.6$ (e) $n = 1$ and (f) $n = 1.4$ for $Ra = 10^5$ and (bottom) (g) $n = 0.6$ (h) $n = 1$ and (i) $n = 1.4$ for $Ra = 10^6$ while $Pr = 100$.

Table 7
The CuLBM simulation times were recorded on both GPU (GPU1: Quadro RTX 6000 with 24 GB GDDR6 RAM and 4608 CUDA cores) and (GPU2:GeForce RTX 4080 SUPER with 16 GB GDDR6X RAM and 10240 CUDA cores) for various lattice configurations of the side-heated cavity problem with the power-law non-Newtonian fluid while $Ra = 10^6$, $Pr = 100$ and $n = 0.6$. The performance comparison is based on 10000 simulation iterations.

Lattice size	Time (s):	GPU1	GPU2	Speed up = t_{GPU1}/t_{GPU2}
128 ×128		0.612964	0.469531	1.31
256×256		2.11917	1.59596	1.33
512×512		7.25527	4.97535	1.46
1024×1024		27.2752	19.6058	1.39
2048×2048		110.457	77.0067	1.44

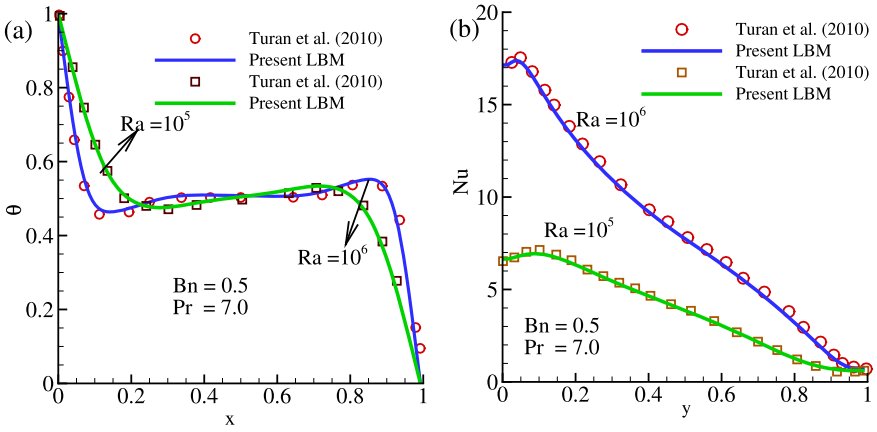


Fig. 14. Comparison of the present results with the results of Turan et al. [42] (a) temperature θ at $y/H = 0.5$ (b) local Nusselt number (Nu) for the different $Ra = 10^5, 10^6$, Bingham number $Bn = 0.5$ while $Pr = 7.0$.

$$x = \frac{\bar{x}}{H}, \quad y = \frac{\bar{y}}{H}, \quad u = \frac{\bar{u}}{u_0}, \quad v = \frac{\bar{v}}{u_0}$$
$$Ra = \frac{g\beta H^3(T_h - T_c)}{\left(\frac{\mu_0}{\rho}\right)\alpha}, \quad P = \frac{\bar{p}}{\rho u_0}, \quad \tau = \frac{\bar{t}u_0}{H} \tag{70}$$
$$\theta = \frac{T - T_c}{T_h - T_c}, \quad Pr = \frac{\left(\frac{\mu_0}{\rho}\right)}{\alpha}, u_0 = \sqrt{g\beta\Delta T H}$$

One of the key non-Newtonian fluids characterized by shear rate and yield stress is the Bingham fluid, where the dimensional viscosity is calculated as follows [42,27]:

$$\mu(|\bar{\gamma}|) = \mu_o + \frac{\tau_s}{|\bar{\gamma}|}\left(1 - \exp\left(-m_0|\bar{\gamma}|\right)\right) \tag{71}$$

where μ_o is the Newtonian viscosity, τ_s is the yield shear stress, m_0 is the growth factor. The non-dimensional form the viscosity of the Bingham fluids is

$$\frac{\mu}{\mu_o} = 1 + \frac{Bn}{|\gamma|}\left(1 - \exp\left(-M_0|\gamma|\right)\right) \tag{72}$$

where $Bn = \frac{\tau_s H}{\mu_0 u_0}$ is the Bingham number, $M_0 = \frac{mu_0}{H}$ is the growth factor which is taken 10000 for the present case.

Figs. 14(a), (b) show a comparison between temperature distributions and the local Nusselt number (Nu) derived from the current Cumulant lattice Boltzmann method (CuLBM) and those reported by Turan et al. [42]. This analysis was conducted at different Rayleigh numbers ($Ra = 10^5$ and $Ra = 10^6$), while keeping the Bingham number ($Bn = 0.5$) and Prandtl number ($Pr = 7.0$) constant. The results from the current LBM are in good agreement with Turan et al. [42] at ($y/H = 0.5$). Interestingly, the temperature distribution is slightly higher in the current LBM at lower Rayleigh numbers, which might indicate a small difference in numerical methods or boundary conditions. However, at higher Rayleigh numbers, the results align more closely, suggesting that the methods are quite robust under these elevated conditions.

As for the local Nusselt number, it tends to decrease as (y) increases for both Rayleigh values. At ($Ra = 10^5$), the Nusselt number shows a gradual decline, following a trend similar to that of Turan et al. [42], but with slightly higher values in the current LBM. When we look at ($Ra = 10^6$), the differences between the two methods become more noticeable, with the current LBM demonstrating a better heat transfer capability. Overall, the results from the current LBM align well with the findings from Turan et al. [42], confirming the effectiveness of the lattice Boltzmann approach in capturing both thermal and fluid dynamics. While there were some moderate discrepancies-especially at lower Rayleigh numbers for temperature and at higher Rayleigh numbers for the Nusselt number-the overall findings suggest that the current LBM method performs comparably. This highlights the potential of LBM for simulating thermal-fluid applications that involve complex flow behavior.

The comparison shown in Table 8 takes a closer look at the average Nusselt number ($\overline{Nu}$) across various Bingham (Bn) and Rayleigh (Ra) numbers, using the current CuLBM method and comparing it to previous studies. Here are some key takeaways: The present CuLBM results are quite similar to those from Turan et al. [42] FVM (for instance, at $Bn = 0.5$: 1.99 compared to 2.00; $Bn = 1.0$: both at 1.70). Huilgol et al. [43] and Kefayati [44] report slightly lower values (like $Bn = 1.5$: Huilgol et al. [43] at 1.49 versus Present at 1.44), which might be due to variations in the solution methodologies (such as SRT-LBM, NS-FEM versus CuLBM) or how yield stress is modeled. The results from Turan et al. [42] and Present CuLBM are in agreement (for example, at $Bn = 2.0$: both at 1.21; $Bn = 3.0$: 1.01 versus 1.10). However, Kefayati [44] and Huilgol [43] show some differences (like $Bn = 2.0$: Kefayati [44] at 1.30 compared to Present at 1.21), indicating possible issues in accurately capturing yield stress effects at higher Bn. The Present CuLBM method tends to produce slightly higher $\overline{Nu}$ values than the others (for instance, at $Bn = 1.0$: 3.94 compared to 3.88-3.89), which could be attributed to better numerical stability in CuLBM. Kefayati's [44] higher $\overline{Nu}$ at $Bn = 3.0$ (3.12 versus Present at 2.96) might be due to differences in how viscoplastic regularization or grid resolution is handled.

Only Turan et al. [42] and Present CuLBM have provided data so far. The results are quite similar (for instance, at $Bn = 5.0$: 2.07 compared to 2.09; and at $Bn = 10$: both are 1.00), which really shows how robust CuLBM is in extreme yield-dominated situations. There are some minor differences at $Bn = 7.0 - 9.0$ (like at $Bn = 9.0$: 1.08 versus 1.02), which might be due to the iterative convergence criteria or the way yield stress is calculated. The strong alignment with Turan et al. [42] (FVM) underscores CuLBM's reliability for both Newtonian and non-Newtonian flows. On the other hand, the discrepancies with Huilgol et al. [43] (NS-FEM) and Kefayati (SRT-LBM) could be due to variations in collision operators, boundary conditions, or the GPU acceleration used in CuLBM. Notably, Huilgol [43] and Kefayati [44] don't have results for $Ra = 10^5$ at $Bn \geq 4$, suggesting their methods might struggle with high Ra and high Bn flows, something that CuLBM handles quite well.

Overall, the Present CuLBM aligns quite well with Turan et al. [42] FVM, particularly at the higher $Ra = 10^5$. The differences noted with Huilgol et al. [43] and Kefayati [44] could stem from the use of different methods (NS-FEM, SRT-LBM versus CuLBM). The present method appears to be quite robust across all Bn values. Additionally, for $Ra = 10^5$, the $\overline{Nu}$ from the Present method is a bit higher than the others at mid Bn, but it tends to converge at the extremes. This could be due to better management of non-Newtonian effects, varying convergence criteria, or enhanced numerical schemes in CuLBM. The agreement at $Bn = 0$ for both Ra indicates that the methods are validated for Newtonian

Table 8
Comparison of the present results in terms of the average Nusselt number ($\overline{Nu}$) with the FVM solution by Turan et al. [42], SRT-LBM solution of Huilgol et al. [43] and Kefayati [44] for the different Bingham number (Bn) while $10^4 \leq Ra \leq 10^5$ and $Pr = 1$.

$Ra = 10^4$	Turan et al. [42]	Huilgol et al. [43]	Kefayati [44]	Present CuLBM
$Bn = 0.$	2.23	2.23	2.23	2.24
$Bn = 0.5$	2.00	1.91	1.95	1.99
$Bn = 1.0$	1.70	1.65	1.68	1.70
$Bn = 1.5$	1.43	1.49	1.45	1.44
$Bn = 2.0$	1.21	1.34	1.30	1.21
$Bn = 2.5$	1.10	1.21	1.18	1.10
$Bn = 3.0$	1.10	1.10	1.05	1.01
$Ra = 10^5$	Turan et al. [42]	Huilgol et al. [43]	Kefayati [44]	Present CuLBM
$Bn = 0$	4.60	4.60	4.60	4.64
$Bn = 1$	3.89	3.89	3.88	3.94
$Bn = 2$	3.45	3.46	3.44	3.49
$Bn = 3$	2.95	3.16	3.12	2.96
$Bn = 4$	2.47	–	–	2.49
$Bn = 5$	2.09	–	–	2.07
$Bn = 6$	1.76	–	–	1.76
$Bn = 7$	1.48	–	–	1.51
$Bn = 8$	1.22	–	–	1.26
$Bn = 9$	1.02	–	–	1.08
$Bn = 10$	1.00	–	–	1.00

Table 9
Comparison of the present results in terms of the average Nusselt number ($\overline{Nu}$) with the FVM solution by Turan et al. [42] for the different Prandtl number (Pr), and Bingham numbers (Bn) while $Ra = 10^6$.

$Ra = 10^6$	Ref. [42] ($Pr = 10$)	Present ($Pr = 10$)	Ref. [42] ($Pr = 100$)	Present ($Pr = 100$)
$Bn = 0$	9.24	9.24	9.26	9.27
$Bn = 1$	7.27	7.26	4.09	4.04
$Bn = 2$	5.60	5.52	1.94	2.01
$Bn = 3$	4.28	4.24	1.18	1.19
$Bn = 4$	3.27	3.24	1.07	1.08
$Bn = 5$	2.57	2.57	1.05	1.05
$Bn = 6$	2.07	1.99	–	–
$Bn = 7$	1.66	1.66	–	–
$Bn = 8$	1.36	1.34	–	–
$Bn = 9$	1.12	1.11	–	–
$Bn = 10$	1.01	1.02	–	–
$Bn = 15$	1.00	1.00	–	–

cases, meaning the differences in non-Newtonian behavior likely arise from how the Bingham model is implemented. Furthermore, the trend of decreasing $\overline{Nu}$ with increasing Bn is consistent across all studies, demonstrating that higher Bingham numbers (which imply more yield stress) lead to reduced heat transfer, aligning with what we would expect from the physics involved. Another compassion is given in Table 9 at the $Ra = 10^6$, considering two Prandtl numbers $Pr = 10, 100$ for the different Bingham numbers Bn. This Table shows that the agreement between the FVM results of Turan et al. [42] and the present CuLBM results is excellent indeed.

The illustration in Fig. 15(a)–(h) provides a side-by-side comparison of the yielded (shown in white) and unyielded (in color) areas for viscoplastic fluids. This comparison highlights the results from our CuLBM approach alongside those from the well-known commercial software, COMSOL Multiphysics 6.3, which relies on the Navier-Stokes equations and the finite element method (NS-FEM). We examined a variety of governing parameters: $Ra = 10^4, 10^5$, $Pr = 0.1, 1, 10$ and the Bingham number $Bn = 1, 3, 18$. The findings clearly indicate that CuLBM effectively captures the unyielded regions, confirming its ability to simulate the behavior of viscoplastic fluids accurately.

Figs. 16(a)-(l) illustrate how the Prandtl number ($Pr = 1, 10, 100, 1000$) affects the streamlines (shown in the left column), isotherms (in the middle column), and unyielded zones (on the right) for a viscoplastic fluid within a side-heated square cavity. This is all at a constant Rayleigh number of $Ra = 10^6$ and a Bingham number of $Bn = 1$. As the Prandtl number increases from 1 to 1000 (as seen in the streamlines in plots a, d, g, j), the central vortex becomes more confined and weaker. When Pr is low at 1, thermal diffusivity is high, which encourages vigorous convection and a stronger recirculation vortex. However, as Pr rises, momentum diffusivity takes precedence over thermal diffusion, leading to a reduction in convective motion and a more localized flow. At the extreme of $Pr = 1000$, the vortex shrinks and weakens, indicating a decrease in convective transport. Looking at the isotherms (plots a, d, g, j) as Pr increases from 1 to 1000, we see that they become increasingly vertical and compressed near the heated wall. With a low Pr, heat spreads more easily, resulting in smoother, curved isotherms throughout the area. In contrast, at high Pr, the thermal boundary layers thin out significantly, suggesting that conduction is limited to the wall area, and heat isn't efficiently distributed across the space. This highlights that fluids with a high Prandtl number have lower thermal conductivity compared to their momentum diffusivity. The unyielded (solid-like) regions, depicted in color (plots c, f, i, l), represent areas where the local stress is below the yield stress, meaning $\tau_y < Bn$. As Pr increases, these unyielded regions expand notably, particularly in the central and corner areas. This expansion is due to the diminished strength of convective flows, which produce less shear stress in high-Pr fluids. By the

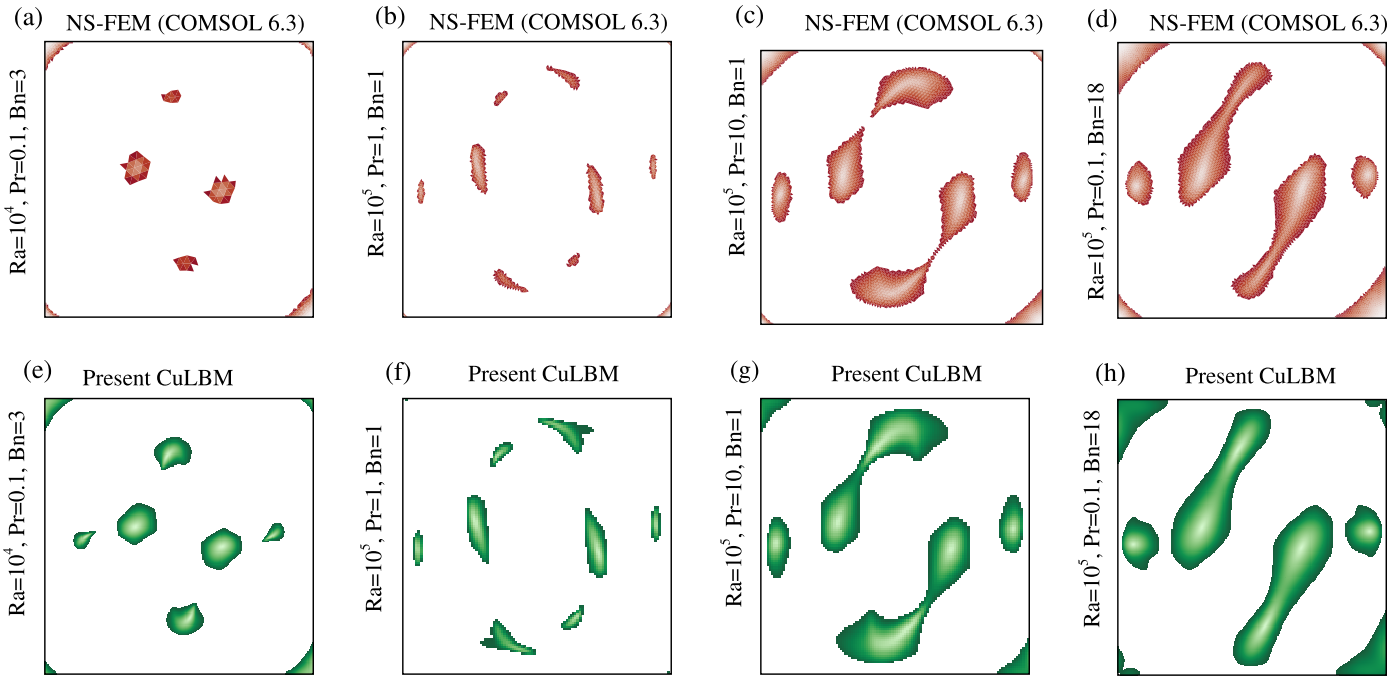


Fig. 15. Unyielded regions are shown in color obtained by the commercial software COMSOL Multiphysics 6.3 based on the Navier-Stokes equations and finite element method (NS-FEM): (a) $Ra = 10^4$, $Pr = 0.1$ and $Bn = 3$ (b) $Ra = 10^5$, $Pr = 1$ and $Bn = 1$ (c) $Ra = 10^5$, $Pr = 10$ and $Bn = 1$ (d) $Ra = 10^5$, $Pr = 0.1$ and $Bn = 18$. The corresponding Unyielded regions are obtained by the present CuLBM: (e) $Ra = 10^4$, $Pr = 0.1$ and $Bn = 3$ (f) $Ra = 10^5$, $Pr = 1$ and $Bn = 1$ (g) $Ra = 10^5$, $Pr = 10$ and $Bn = 1$ (h) $Ra = 10^5$, $Pr = 0.1$ and $Bn = 18$.

Table 10
The MRT-LBM and CuLBM simulation performance analysis on GPU1: Quadro RTX 6000 with 24 GB RAM and 4608 CUDA cores is tested for various lattice configurations of the side-heated cavity problem with the non-Newtonian Bingham fluid while $Ra = 10^6$, $Pr = 1000$ and $Bn = 1$. The performance comparison is based on 10000 simulation iterations.

Lattice size	Time (s) for:	MRT-LBM	CuLBM	Speed = MRT-LBM/CuLBM
128 ×128		0.688272	0.672524	1.02
256×256		0.707533	2.33533	0.31
512×512		2.5397	8.21974	0.31
1024×1024		8.26152	32.2077	0.26
2048×2048		32.4571	135.26	0.24

Table 11
The MRT-LBM and CuLBM simulation performance analysis on GPU2 (GeForce RTX 4080 with 16 GB RAM and 10240 CUDA cores) is tested for various lattice configurations of the side-heated cavity problem with the non-Newtonian Bingham fluid while $Ra = 10^6$, $Pr = 1000$ and $Bn = 1$. The performance comparison is based on 10000 simulation iterations.

Lattice size	Time (s) for:	MRT-LBM	CuLBM	Speed = MRT-LBM/CuLBM
128 ×128		0.472123	0.445399	1.06
256×256		0.47736	1.51904	0.31
512×512		1.51097	4.71581	0.32
1024×1024		4.84459	18.5727	0.26
2048×2048		18.474	72.8137	0.25

time we reach $Pr = 1000$ (as shown in plot l), a large portion of the domain is unyielded, indicating that the flow is largely hindered by yield stress. The way the yielded and unyielded distribution varies with Pr.

Tables 10 and 11 provide a comparison of the simulation performance—specifically time and speedup—between the MRT-LBM and Cumulant LBM (CuLBM) methods for a non-Newtonian Bingham fluid in a side-heated cavity problem, utilizing two different GPUs with parameters set at $Ra = 10^6$, $Pr = 1000$, and $Bn = 1$. When working with smaller grid sizes (128×128), CuLBM shows a slight edge in speed over MRT-LBM. However, as the increase of the grid size, CuLBM starts to lag behind, and MRT-LBM takes the lead. This performance gap becomes more pronounced with larger grid sizes, suggesting that CuLBM struggles with scalability on GPU1. While the performance trend remains consistent for GPU1, CuLBM actually performs better on GPU2, thanks to its higher CUDA core count and more advanced architecture. MRT-LBM is more cost-effective computationally, as it relies on simpler transformation matrices and fewer floating-point operations. On the other hand, CuLBM, while offering greater stability and accuracy-especially in scenarios with high Reynolds numbers (Re) or Rayleigh numbers (Ra)-requires complex cumulant transformations that are quite demanding

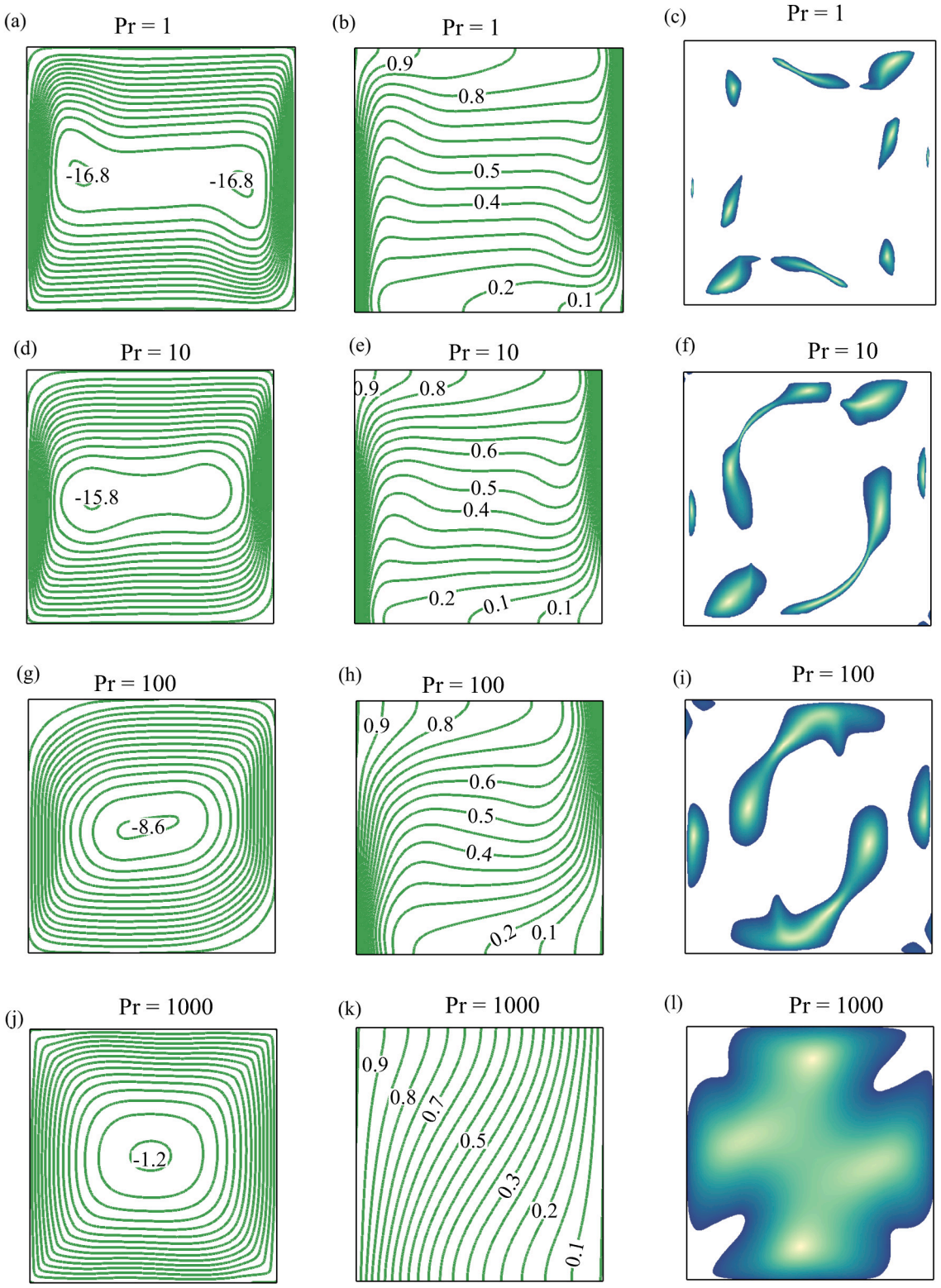


Fig. 16. Influence of the Prandtl number (Pr) on the streamlines (left), isotherms (middle) and unyielded zones (right): (a), (b), (c) for $Pr = 1$, (d), (e), (f) for $Pr = 10$, (g), (h), (i) for $Pr = 100$ and (j), (k), (l) for $Pr = 1000$ while $Ra = 10^6$, and $Bn = 1$.

computationally. It has been run a series of comparative simulations where the standard MRT model failed to converge or showed instability at high $Ra \geq 10^9$ under the same grid and viscosity conditions (512×512) with $\nu = 0.005$. In contrast, CuLBM maintained stability and provided a good approximation to the benchmark results. Thus, for problems where accuracy is critical, especially at larger Re or Ra, CuLBM is the preferred choice.

10. Conclusion

In conclusion, a new two-dimensional cumulant-based lattice Boltzmann method (CuLBM) has been effectively developed using the D2Q9 lattice model for velocity and the D2Q5 lattice model for temperature. This implementation, optimized for GPU computing with CUDA C/C++, has been validated against natural convection in a side-heated square cavity across a broad range of Rayleigh numbers ($Ra = 10^3$ to 10^9) for the Newtonian fluids and the maximum $Ra = 10^6$ for the non-Newtonian fluids. The results, which include streamlines, isotherms, velocity, temperature, and entropy generation, demonstrate the accuracy and reliability of the proposed CuLBM. The successful replication of published benchmark data really confirms that the CuLBM code structure is on point, along with its implementation of power-law viscosity based shear-thinning ($n < 1$) and shear-thickening ($n > 1$) fluids.

The present CuLBM is capable of replicating the side heated square cavity problem for the maximum $Ra = 10^9$. The relative percentage errors between the CuLBM results and the reference values are impressively low, typically remaining under 1% when the Rayleigh number is $Ra = 10^7$, and even dropping below 0.5% at the same Rayleigh number.

It has been conducted a series of comparative simulations where the standard MRT-LBM model struggled to converge or became unstable at high $Ra = 10^9$, all while using the same grid and viscosity settings (512×512) with $\nu = 0.005$. On the other hand, CuLBM held its ground, maintaining stability and delivering a solid approximation to the benchmark results. Therefore, when accuracy is paramount-especially at higher Re or Ra, the CuLBM stands out as the go-to option. This study also highlighted how GPU performance is heavily influenced by grid size, showing that larger lattices can achieve significantly better speed and computational efficiency compared to their smaller counterparts when measured against CPU performance.

The Present CuLBM shows a great match with the established FVM results and expands its validity into the moderate Ra and high Bingham numbers Bn. Some small differences in the mid-Bn ranges suggest that we need to take a closer look at the numerical methods and yield stress modeling. All in all, CuLBM demonstrates its strength in simulating Bingham fluids across a wide range of parameters. The Prandtl number is essential for understanding how flow and heat transfer work in viscoplastic fluids. When the Prandtl number is low, it encourages strong convection and creates more fluid-like regions. On the other hand, a high Prandtl number dampens convection, limits heat diffusion to areas close to the walls, and results in larger solid-like zones. These insights are crucial for grasping heat transfer behavior in non-Newtonian fluids, such as Bingham plastics, which are commonly used in various industrial processes like food production, polymer manufacturing, and drilling fluids. The findings clearly indicate that CuLBM effectively captures the unyielded regions, confirming its ability to simulate the behavior of viscoplastic fluids accurately.

The validation against established results really boosts our confidence in using LBM for predictive modeling, especially in practical applications such as heat exchangers and thermal management systems. These validations also boost our confidence in using the current solver for tackling more complex geometries by considering the curved boundary treatment and diving into three-dimensional simulations in future research. With its enhanced performance compared to traditional BGK, MRT-LBM and the finite volume methods, the developed 2D CuLBM stands out as a powerful tool for efficiently addressing convective heat transfer challenges.

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Declaration of competing interest

The author declares the following financial and personal relationships that may be perceived as potential competing interests: Md. Mamun Molla is employed by North South University. However, there is no conflict of interest with the funding agency or any other individual.

Data availability

Data will be made available on request.

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